

Uniform Transitions and Integral Structures in Polylogarithmic Analysis

A research continuation of the ProveIt manuscript

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Abstract

We extend three parts of the ProveIt polylogarithms programme and reconcile an outstanding pair of weight-seven candidates. For alternating harmonic sums, a rescaled saddle and a global gamma-phase majorant give a relative expansion at every fixed order, uniform for all depths and all positive shifts as the outer exponent grows. We invert the first-pole transition and obtain explicit logarithmic corrections to its threshold. For reflected log-gamma moments with both exponents growing, we prove a nontrivial transition factor, its first two corrections, and exact coefficient identities at every order. An exact interval certificate also proves global monotonicity of a reflected log-gamma slope ratio, giving a unique nondegenerate saddle and a full expansion for every fixed positive ratio of the two exponents. For weighted distribution relations, an explicit integral grid basis supports a complete two-prime primitive-cokernel calculation, including torsion, modular ranks, Smith factors, and optimal rational reduction denominators. Explicit level-fifteen Hurwitz-zeta identities follow. Two convergent shuffles prove that the two recorded S_6 candidates are equivalent; the common evaluation remains conjectural. Full proofs, exact verifiers, clearly classified numerical diagnostics, source receipts, editorial corrections, and a further research programme accompany the article.

Keywords. Multiple polylogarithms; alternating harmonic sums; Laplace asymptotics; gamma moments; universal distributions; Smith normal form; Hurwitz zeta; exact certificates.

Baseline. Immutable repository commit `cc34f73596336f2466d9754cb0f3635bd2bedade`, including the six incoming archives deposited with it. Results are distinguished from that existing material throughout.

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1 Scope, conventions, and principal results

This article continues the research programme in *Polylogarithms and their Arithmetic Bridges* at the immutable ProveIt commit `cc34f73596336f2466d9754cb0f3635bd2bedade`, dated 10 October 2026 [1]. The baseline includes both the consolidated manuscript and six newly deposited research archives. This distinction matters: several conjectures still labelled open in the consolidated text already have proofs in those incoming archives. The present contribution does not count those proofs as new work. An accompanying inventory records the exact source and archive hashes.

The results concern three structures which recur throughout the programme. Positive Mellin integrals control alternating harmonic sums; gamma concentration describes a transition in reflected log-gamma moments; integral distribution relations organize rational-grid special values. A shorter fourth part reconciles the two recorded weight-seven Gaussian candidates by an exact shuffle identity. The proofs are analytic and algebraic, with one explicitly identified computer-assisted inequality whose finite middle-interval step uses exact integer arithmetic. Other computations supply reproducible checks and examples, and are identified explicitly when they provide only numerical evidence.

1.1 What is already known in the pinned material

The consolidated manuscript already proves the mixed Gaussian S_4 evaluation, signed first-omitted-term bounds for the harmonic family, its first-pole transition, an all-orders expansion within that transition, and a saddle expansion when the exponent-to-depth ratio stays in a fixed positive compact interval. Its reflected log-gamma theory includes fixed-reflected-exponent expansions, late coefficients, and a sharp optimal-truncation remainder. Its distribution theory includes arbitrary complex weights, conductor coordinates, exceptional spectral jets, and a primitive-grid determinant up to a constant from the choice of complex coordinates.

The six incoming archives additionally include proofs of the odd-weight rank conjecture for the signed depth-two product module, extensions to other levels, integral Smith calculations for that product module, independent S_4 certificates, complement-collision classifications, CM evaluations, and further zero-geometry results. These are distinct from the integral distribution lattice studied here. A small apparent coefficient relation found numerically at a higher weight is not made a theorem by being stored in an incoming report. In particular, both recorded S_6 formulas remain conjectural.

1.2 The contributions proved here

1. **A single harmonic saddle over the whole moving range.** Theorem 2.1 gives an all-orders *relative* expansion as $p \rightarrow \infty$, uniformly for all integers $h \geq 1$ and all $a > 0$. It allows p/h to tend to zero, to remain bounded, or to grow without bound. The exact saddle incorporates the shift a . A global gamma-phase upper bound makes this uniformity possible. This resolves the request in the proportional-depth chapter for an approximation which continues through the growing-ratio transition.
2. **An inverse transition with explicit corrections.** Theorem 3.2 determines the unique real outer exponent at which the normalized harmonic sum reaches a prescribed value $\theta \in (0, 1)$. It gives a finite recursion to every order and prints the first two correction polynomials. Its error is proved by asymptotic brackets, without differentiating an unspecified remainder.
3. **A transition when both reflected moment exponents grow.** Theorem 4.1 treats $(n + 1)/(m + 1) = \log m + O(1)$ with $m \rightarrow \infty$. The fixed- m endpoint formula acquires a surviving factor

$$\exp \left[\left(\frac{\zeta(2)}{2\gamma} - \gamma \right) m \exp \left(-\frac{n+1}{m+1} \right) \right].$$

Two corrections and an all-orders coefficient rule are proved. Corollary 4.5 gives exact identities for two extremal coefficient families in every order.

4. **A unique saddle for every proportional moment.** Theorem 5.1 proves that

$$x \mapsto \frac{-\psi(x) \log \Gamma(1-x)}{-\psi(1-x) \log \Gamma(x)}$$

is strictly increasing on $(0, 1)$. The proof combines analytic endpoint estimates with a small exact rational interval certificate. Theorem 5.3 consequently gives the complete moment expansion for exponent ratios in any compact positive interval, including a unique nondegenerate saddle. This settles another part of the simultaneous large-exponent problem.

5. **Integral coordinates and the complete two-prime primitive cokernel.** Theorem 6.1 gives a concrete grid basis over every commutative coefficient ring. Its finite resolution is an explicit adaptation of classical universal-distribution methods. The principal additional calculation is Theorem 6.5: a complete presentation of the primitive-grid cokernel at $q = p\ell$, including torsion at singular weights. Its Smith factors give the exact denominator required to express all grid points in primitive coordinates. Explicit level-fifteen Hurwitz-zeta identities illustrate the result.

6. **An exact reconciliation of the weight-seven candidates.** Proposition 7.1 proves a coordinate identity among Gaussian doubles and shows that the two S_6 candidates are the same conjecture. The displayed certificate uses two convergent shuffles. An incomplete larger relation search is reported as inconclusive.

The phrase “additional result” means an extension of the audited baseline, with the antecedents stated below. It is not a claim of global priority after an exhaustive literature search. In particular, the freeness/resolution principle in universal distributions is classical; its role here is to support explicit integral calculations, not to rename an existing general theory.

1.3 Notation and interpretation

We use γ for Euler’s constant and $\beta(s) = \sum_{n \geq 0} (-1)^n (2n+1)^{-s}$ for the Dirichlet beta function when the defining series converges. The multiple-polylogarithm convention is

$$\text{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}},$$

continued only where a continuation is expressly indicated. All Gaussian shuffles used for the coordinate theorem converge ordinarily. The normalized harmonic quantity $R_{p,h}$ and the normalized moment $\mathcal{R}_{n,m}$ are different functions. The symbol λ is a local transition parameter defined separately in each section.

The term *formal distribution module* refers to symbols subject to specified linear multiplication relations. Its rank is not the dimension of the span of the corresponding evaluated special values. Similarly, a Smith factor describes a particular integral inclusion; it is not an assertion of algebraic or transcendence independence. These distinctions are essential at exceptional weights and in finite characteristic.

1.4 Reading and integration guide

Sections 2–3 extend the harmonic chapters. Sections 4–5 belong after the fixed- m reflected-moment and sharp-remainder material. Section 6 supplements the distribution and conductor chapters. Section 7 belongs beside the weight-seven conjecture. Section 8 records proposed

corrections and status changes; the final sections describe the computational evidence and a prioritized research agenda. The archive includes both an assembled $\text{\TeX}$ article and modular section files, together with exact verifiers, numerical diagnostics, source receipts, and a small proposed editorial patch.

2 A single harmonic saddle throughout the moving range

For an integer $h \geq 1$ define

$$e_h(n) = [u^h] \prod_{j=1}^n (1 + u/j), \quad T_{p,h}(a) = \sum_{n \geq 0} \frac{(-1)^n e_h(n)}{(n+a)^p}, \quad p, a > 0.$$

The empty product is one, so $e_h(n) = 0$ for $n < h$ and $e_h(h) = 1/h!$. In particular $e_1(n) = H_n$. The first nonzero term of the alternating series motivates the normalization

$$R_{p,h}(a) = (-1)^h h! (h+a)^p T_{p,h}(a). \quad (2.1)$$

For $a = 1/2$, the Gaussian harmonic family in the manuscript is $S_{p,h} = 2^{-p} T_{p,h}(1/2)$; its usual S_p means $S_{p,1}$.

2.1 The exact positive integral

We briefly justify the starting point to keep the asymptotic argument self-contained. The generating function is

$$\sum_{n \geq 0} e_h(n) z^n = \frac{(-\log(1-z))^h}{h!(1-z)}, \quad |z| < 1. \quad (2.2)$$

It follows by taking the coefficient of u^h in $\sum_{n \geq 0} \prod_{j=1}^n (1 + u/j) z^n = (1-z)^{-1-u}$. Ordinary convergence of $T_{p,h}$ for $p > 0$ can also be checked directly. The recurrence $e_h(n) - e_h(n-1) = e_{h-1}(n-1)/n$ and $e_h(n) = O_h((1 + \log n)^h)$ show that $e_h(n)/(n+a)^p \rightarrow 0$ and its successive differences have an absolutely summable majorant of order $n^{-p-1}(1 + \log n)^h$. Summation by parts against $(-1)^n$ therefore proves convergence. Absolute convergence holds for $p > 1$.

Insert an Abel factor $0 < \rho < 1$ in the sum and use $(n+a)^{-p} = \Gamma(p)^{-1} \int_0^\infty y^{p-1} e^{-(n+a)y} dy$. Equation (2.2) and dominated convergence as $\rho \uparrow 1$ give

$$R_{p,h}(a) = \frac{(h+a)^p}{\Gamma(p)} \int_0^\infty \frac{y^{p-1} e^{-ay} L(y)^h}{1 + e^{-y}} dy, \quad L(y) = \log(1 + e^{-y}). \quad (2.3)$$

The bound $L(y) \leq \min(\log 2, e^{-y})$ controls both endpoints. This integral, including its positivity, is already established in the manuscript [1, chapters/04-depth.tex, chapters/04-proportional-depth.tex].

Define

$$k_h(t) = \frac{1}{1+t} \left(\frac{\log(1+t)}{t} \right)^h, \quad k_h(0) = 1.$$

If Y_p has the gamma law with shape p and rate $h+a$, then

$$R_{p,h}(a) = \mathbb{E} k_h(e^{-Y_p}). \quad (2.4)$$

The kernel k_h is strictly decreasing on $[0, 1]$, with $k_h(1) = (\log 2)^h/2$ and $k_h(0) = 1$. Consequently

$$\frac{(\log 2)^h}{2} < R_{p,h}(a) < 1. \quad (2.5)$$

2.2 The moving saddle and the uniform expansion

Put

$$w(y) = -\frac{d}{dy} \log L(y) = \frac{1}{(1+e^y)L(y)}, \quad \alpha = p/h, \quad \epsilon = a/h.$$

Writing $v = e^{-y}$ gives

$$\frac{d}{dv} \log \frac{v}{(1+v) \log(1+v)} = \frac{\log(1+v) - v}{v(1+v) \log(1+v)} < 0.$$

Thus w increases strictly from $w_0 = 1/(2 \log 2)$ to 1. There is exactly one $x > 0$ satisfying

$$x(w(x) + \epsilon) = \alpha, \quad \frac{p}{h+a} < x < \frac{p}{hw_0+a}. \quad (2.6)$$

This is the saddle for the full exponent after absorbing the factor e^{-ay} into it. Keeping a in the exponent is useful when a itself varies.

At this x set

$$B = 1 + \frac{x^2 w'(x)}{\alpha}, \quad A(z) = \frac{1}{z(1+e^{-xz})}, \quad (2.7)$$

$$\Psi(z) = \log z + \frac{\log L(xz) - \log L(x) - \epsilon x(z-1)}{\alpha}. \quad (2.8)$$

Then $\Psi(1) = \Psi'(1) = 0$ and $\Psi''(1) = -B$.

Theorem 2.1 (Uniform expansion with no ratio restriction). *For each fixed integer $J \geq 0$, as $p \rightarrow \infty$, uniformly over all integers $h \geq 1$ and all real $a > 0$,*

$$R_{p,h}(a) = \mathcal{S}_{p,h}(a) \left\{ \sum_{j=0}^J \frac{E_j(x, \epsilon)}{p^j} + O_J(p^{-J-1}) \right\}, \quad (2.9)$$

where $E_0 = 1$ and

$$\mathcal{S}_{p,h}(a) = \frac{((h+a)x)^p e^{-ax} L(x)^h}{\Gamma(p)(1+e^{-x})} \sqrt{\frac{2\pi}{pB}}. \quad (2.10)$$

Every coefficient E_j is bounded on the entire parameter range. In particular, with all derivatives evaluated at $z = 1$,

$$E_1 = \frac{A''}{2BA} + \frac{\Psi''' A'}{2B^2 A} + \frac{\Psi''''}{8B^2} + \frac{5(\Psi''')^2}{24B^3}. \quad (2.11)$$

The constants in the relative remainder depend on J alone.

Proof. Substitute $y = xz$ in (2.3). After removing its value at the saddle, the remaining integral is

$$\int_0^\infty e^{p\Psi(z)} A(z) dz. \quad (2.12)$$

We verify the uniform conditions for Laplace expansion directly. The local expansion itself is the usual one [5, §2.3(iii)]; the global parameter control is the point requiring proof here.

First, the concavity of $\log L$, equivalent to $w' > 0$, gives its tangent bound at x . Combining this bound with the saddle equation yields the parameter-free majorant

$$\boxed{\Psi(z) \leq \log z - z + 1 \quad (z > 0).} \quad (2.13)$$

Also $A(z) \leq z^{-1}$. For each fixed $0 < d < 1$, the contribution of $|z-1| \geq d$ to (2.12) is bounded by the corresponding tail of $e^p \int_0^\infty z^{p-1} e^{-pz} dz$. The gamma Chernoff bound, followed by Stirling's formula, bounds that tail by $C_d p^{-1/2} e^{-cdp}$ for positive constants independent of h and a .

Second, the saddle curvature is bounded above and below uniformly. For $r_x = (1 + e^x)^{-1}$, logarithmic differentiation gives $w' = w(w - 1 + r_x)$, hence $0 < w' \leq r_x \leq e^{-x}$. Therefore

$$1 \leq B = 1 + \frac{xw'(x)}{w(x) + \epsilon} \leq 1 + \frac{2 \log 2}{e}, \quad \frac{1}{2} \leq A(1) \leq 1. \quad (2.14)$$

For z in any fixed closed neighborhood of 1 contained in $(0, \infty)$, every fixed derivative of A is uniformly bounded. For $k \geq 2$, the nonlogarithmic part of $\Psi^{(k)}(z)$ has absolute value

$$\frac{x^k |w^{(k-1)}(xz)|}{\alpha} = \frac{x^{k-1} |w^{(k-1)}(xz)|}{w(x) + \epsilon}. \quad (2.15)$$

Near $x = 0$, the analyticity of w and $w \geq w_0$ bound this expression. As $x \rightarrow \infty$, use $w^{(j)}(u) = O_j(e^{-u})$ for $j \geq 1$ and the boundedness of $x^j e^{-cx}$ for fixed j and $c > 0$. These observations also prove the required bounds for the logistic amplitude. The linear ϵ term has disappeared from all derivatives of order at least two; increasing ϵ only increases the denominator in (2.15). This explains why no upper bound on a is needed.

Finally, $\Psi''(z) = -z^{-2} - x^2 w'(xz)/\alpha \leq -z^{-2}$. On the fixed saddle neighborhood this supplies a common Gaussian bound after $z = 1 + u/\sqrt{p}$. To any fixed order one may Taylor expand the phase and amplitude there, retaining enough derivatives to bound the remainder by p^{-J-1} times a fixed integrable polynomial times a Gaussian. One explicit justification is to first restrict to $|u| \leq p^\nu$, with a sufficiently small positive ν depending on the desired order, expand to a higher finite Taylor order, and use the common Gaussian bound on its complement. All derivative bounds and all choices are independent of the parameters. Extending each retained Gaussian monomial to the real line incurs an exponentially small error. Odd monomials integrate to zero. The even moments give

$$\int_0^\infty e^{p\Psi(z)} A(z) dz = A(1) \sqrt{\frac{2\pi}{pB}} \left\{ \sum_{j=0}^J E_j p^{-j} + O_J(p^{-J-1}) \right\}.$$

Restoring the exact prefactor proves (2.9). Taking the four possible even Gaussian contributions at order p^{-1} gives (2.11). $\square$

There is a finite prescription for every coefficient. Let $\mathbb{E}_B$ denote expectation under a centered normal law of variance $1/B$, and let τ be formal. Then

$$E_j = \frac{1}{A(1)} [\tau^{2j}] \mathbb{E}_B \left[A(1 + \tau u) \exp \left(\sum_{k \geq 3} \frac{\Psi^{(k)}(1)}{k!} u^k \tau^{k-2} \right) \right]. \quad (2.16)$$

Coefficient extraction precedes Gaussian integration. Only finitely many derivatives and moments enter, so this is an exact algorithm for any prescribed order, not an assertion that the complete asymptotic series converges.

2.3 Evaluation, overlaps, and limitations

For the first correction, useful formulas are

$$\frac{A'(1)}{A(1)} = -1 + x r_x, \quad \frac{A''(1)}{A(1)} = 2 - 2x r_x + x^2 (2r_x^2 - r_x), \quad (2.17)$$

$$\Psi'''(1) = 2 - \frac{x^3 w''(x)}{\alpha}, \quad \Psi''''(1) = -6 - \frac{x^4 w'''(x)}{\alpha}. \quad (2.18)$$

As $x \rightarrow 0$ or $x \rightarrow \infty$, $E_1 \rightarrow 1/12$ uniformly in $\epsilon \geq 0$. In the genuine first-pole limit, for example fixed h, a and $p \rightarrow \infty$, this correction cancels the first reciprocal Stirling correction in the

prefactor, consistently with $R \rightarrow 1$. The condition $x \rightarrow \infty$ alone is insufficient for that conclusion when h grows. For instance $p/h = \frac{1}{2} \log h$, with fixed a , still lies below the first-pole threshold and gives $R \rightarrow 0$.

The theorem includes several formerly separate ranges: $h = p^3$ gives $p/h \rightarrow 0$; $h \asymp p$ gives a proportional ratio; $p \asymp h \log h$ gives the first-pole transition; and fixed h gives an unbounded ratio. It also permits shifts such as $a = p^2$. It does not cover a bounded outer exponent p . That regime retains the separate endpoint expansion in the manuscript. Nor does an O_J theorem itself supply a certified numerical constant at a specified small p .

For numerical work, compute $\log \mathcal{S}$ and standardize around the saddle, rather than directly forming quantities which may be exponentially small. The accompanying program uses $u = \sqrt{pB}(z - 1)$ in the exact integral. Table 1 reports selected checks; its last column is the absolute discrepancy in the normalized bracket, $|R/\mathcal{S} - 1 - E_1/p|$.

Table 1: Diagnostics for the uniformly scaled saddle formula. Values come from 50-digit quadrature and are not interval bounds.

p	h	a	$ R/\mathcal{S} - 1 $	$ R/\mathcal{S} - 1 - E_1/p $
40	64000	1/2	$2.08546 \cdot 10^{-3}$	$2.12816 \cdot 10^{-6}$
160	4096000	1/2	$5.20968 \cdot 10^{-4}$	$1.34979 \cdot 10^{-7}$
40	40	1/2	$4.91662 \cdot 10^{-4}$	$1.05027 \cdot 10^{-5}$
160	160	1/2	$1.15818 \cdot 10^{-4}$	$6.80536 \cdot 10^{-7}$
160	1	1/2	$5.20968 \cdot 10^{-4}$	$1.34979 \cdot 10^{-7}$
40	40	1600	$2.08539 \cdot 10^{-3}$	$2.12726 \cdot 10^{-6}$

At $(p, h, a) = (160, 4096000, 1/2)$ the logarithm of the leading prefactor is about $-1.501185 \cdot 10^6$. Relative asymptotics remain informative even there. The finite test set is a check on signs, normalizations, and the predicted order of error; the parameter-wide statement follows from the proof, not from these samples.

3 Reversing the first-pole transition

The forward transition answers how close an alternating harmonic sum is to its first term. The inverse question is equally natural: how large must the real outer exponent be to attain a specified normalized fraction? The monotonicity needed to pose this question is exact.

Lemma 3.1 (Strict order in the outer exponent). *For fixed $h \geq 1$ and $a > 0$, the function $p \mapsto R_{p,h}(a)$ is continuous and strictly increasing on $(0, \infty)$, with endpoint limits $(\log 2)^h/2$ and 1.*

Proof. For $d > 0$, independent gamma variables with common rate $h + a$ can be coupled by $Y_{p+d} = Y_p + Y_d$. Since $y \mapsto k_h(e^{-y})$ is strictly increasing and $Y_d > 0$ almost surely, (2.4) proves strict increase. As $p \downarrow 0$, $Y_p \rightarrow 0$ in probability, for instance by its mean. As $p \rightarrow \infty$, $Y_p \rightarrow \infty$ in probability by its mean and variance. The bounded continuous kernel gives the two limits. The same gamma representation, or dominated convergence in (2.3) on compact p -intervals, gives continuity. $\square$

For $0 < \theta < 1$ put

$$\lambda = -\log \theta, \quad L_0 = \log h - \log(2\lambda). \quad (3.1)$$

For all sufficiently large h , Lemma 3.1 therefore gives a unique $p_\theta(h, a) > 0$ satisfying $R_{p_\theta, h}(a) = \theta$.

Theorem 3.2 (All-orders inverse transition). *Fix compact intervals of a in $(0, \infty)$ and of θ in $(0, 1)$. For every fixed integer $J \geq 0$,*

$$p_\theta(h, a) = hL_0 + \sum_{j=0}^J \frac{Q_j(L_0; \lambda, a)}{h^j} + O_J\left(\frac{L_0^{J+2}}{h^{J+1}}\right), \quad h \rightarrow \infty, \quad (3.2)$$

uniformly on those intervals. Here $Q_j \in \mathbb{Q}[L_0, \lambda, a]$ has degree at most $j+1$ in L_0 . The first two polynomials are

$$Q_0(L; \lambda, a) = \left(a + \frac{1}{2} - \frac{\lambda}{2}\right) L + 2 - \frac{5\lambda}{6}, \quad (3.3)$$

$$Q_1(L; \lambda, a) = \left(\frac{\lambda^2}{8} - \frac{\lambda}{4}\right) L^2 + \left(\frac{\lambda^2}{3} - \frac{\lambda}{3} - \frac{1}{12}\right) L - \frac{5a\lambda}{6} + 2a + \frac{3\lambda^2}{8} - \frac{\lambda}{12} - 1. \quad (3.4)$$

3.1 The forward coefficients used in the inversion

We state the input with its provenance, since it is already proved in the baseline. Write $L = p/h$ and $v = he^{-L}/2$. In the compact transition window $L = \log h + O(1)$, the all-orders compiler of [1, chapters/04-depth.tex, Theorem `harmonic:thm:allorders`] gives

$$R_{p,h}(a) = e^{-v} \sum_{j=0}^N \frac{\Theta_j(L, v; a)}{h^j} + O_N\left(\frac{L^{N+1}}{h^{N+1}}\right). \quad (3.5)$$

Each Θ_j has degree at most j in L , and

$$\Theta_0 = 1, \quad \Theta_1 = \left(\frac{v^2}{2} - (a + \frac{1}{2})v\right) L + \frac{5}{6}v^2 - 2v. \quad (3.6)$$

A finite way to regenerate these polynomials is as follows. Set

$$\mathcal{D}P(v) = vP(v) - vP'(v), \quad d_j(t; a) = \frac{(-1)^j}{j+1} \{a^{j+1} - (a-t)^{j+1}\}.$$

Define $\mathcal{U}_j$ by the formal kernel expansion

$$k_h(2v/h) = e^{-v} \sum_{j \geq 0} \mathcal{U}_j(v) h^{-j}.$$

Then, as a formal power series in z ,

$$\sum_{j \geq 0} \Theta_j(L, v; a) z^j = \exp\left(L \sum_{k \geq 1} z^k d_k(\mathcal{D}; a)\right) \sum_{k \geq 0} \mathcal{U}_k(v) z^k. \quad (3.7)$$

All operations truncate after finitely many terms at a requested order. For example $\mathcal{U}_1 = 5v^2/6 - 2v$ and $\mathcal{U}_2 = 25v^4/72 - 8v^3/3 + 4v^2$. The exact gamma cumulant identity behind the compiler is

$$\log \mathbb{E} e^{t(Y_p - L)} = -Lt + hL \{\log(1 + a/h) - \log(1 + (a-t)/h)\}.$$

Thus the forward coefficient rule is algebraic, and its uniform error comes from bounded kernel derivatives and centered gamma moments.

3.2 Formal inversion and a rigorous error bracket

Proof of Theorem 3.2. Set $z = 1/h$ and write $p/h = L_0 + \delta$, so $v = \lambda e^{-\delta}$. The formal equation for the inverse is

$$0 = \lambda - \lambda e^{-\delta} + \log \left(\sum_{j \geq 0} z^j \Theta_j(L_0 + \delta, \lambda e^{-\delta}; a) \right). \quad (3.8)$$

It has a unique solution $\delta = \sum_{j \geq 0} z^{j+1} Q_j(L_0; \lambda, a)$ because its coefficient of δ at $z = 0, \delta = 0$ is $\lambda \neq 0$. The recursion is effective: substitute known coefficients and divide the next residual coefficient by $-\lambda$.

There are no hidden denominators in λ . For $j > 0$ every $\Theta_j(L, v; a)$ is divisible by v : this follows directly from (3.7), since $\mathcal{D}$ preserves the ideal (v) , $d_k(0; a) = 0$, and $\mathcal{U}_j(0) = 0$ for $j > 0$. The logarithm in (3.8) is therefore coefficientwise divisible by λ . After division the derivative with respect to δ is one. Induction gives $Q_j \in \mathbb{Q}[L_0, \lambda, a]$. The bound on the degree in L_0 follows by the same induction: a term of order z^r has degree at most r in L_0 , including products and the substitutions in (3.8).

At order z , the equation is $\lambda Q_0 + \Theta_1(L_0, \lambda; a) = 0$, proving (3.3). At order z^2 it gives

$$Q_1 = \frac{Q_0^2}{2} - \frac{Q_0(\partial_L - \lambda \partial_\lambda) \Theta_1 + \Theta_2 - \Theta_1^2/2}{\lambda},$$

evaluated at $(L_0, \lambda; a)$. Expanding the forward compiler through order two gives (3.4). The accompanying exact program checks these cancellations both in the logarithmic equation and in the original normalized expression. A second program derives the same answer directly from gamma cumulants after shifting p .

For the analytic error, let $p_* = hL_0 + \sum_{j=0}^J h^{-j} Q_j$ and use (3.5) through order $J+1$. The formal cancellations prove

$$R_{p_*, h}(a) - \theta = O(h^{-J-2} L_0^{J+2})$$

uniformly on the stated compact sets. Apply the same finite expansion at $p_* \pm Kh^{-J-1} L_0^{J+2}$, with $K > 0$ fixed. The leading term changes by

$$\pm \theta \lambda K h^{-J-2} L_0^{J+2} (1 + o(1)),$$

and the change of the finitely many correction terms is of smaller relative order. Since $\theta \lambda$ is bounded away from zero on a compact subinterval of $(0, 1)$, taking K sufficiently large places the two expanded values on opposite sides of θ . Both exponents remain in the same compact transition window. Lemma 3.1 then brackets the unique inverse between them, proving (3.2). No derivative of the remainder in (3.5) has been assumed. $\square$

3.3 A quantitative example

For $\theta = 1/2$ and $a = 1/2$, numerical inversion of the exact positive integral gives the values in Table 2. The final three columns report absolute exponent errors, successively using hL_0 , $hL_0 + Q_0$, and $hL_0 + Q_0 + Q_1/h$. These are high-precision diagnostics of the defining integral, not interval-certified root enclosures.

Table 2: The inverse harmonic transition at $\theta = a = 1/2$.

h	numerical $p_{1/2}$	leading error	with Q_0	with Q_1/h
50	182.994297936847	3.72486	$4.03044 \cdot 10^{-2}$	$3.19479 \cdot 10^{-3}$
200	998.989394784027	4.65277	$1.82329 \cdot 10^{-2}$	$4.26953 \cdot 10^{-4}$
800	5091.951714446499	5.56974	$7.10744 \cdot 10^{-3}$	$4.78128 \cdot 10^{-5}$

The increasing error of the leading approximation is expected: Q_0 contains a multiple of $\log h$. The decreasing errors after correction illustrate why solving only the leading transition profile can be inadequate even when the forward normalized sum is well approximated.

4 A uniform transition for two large reflected log-gamma exponents

Write

$$f(x) = \log \Gamma(x), \quad M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx, \quad n, m \in \mathbb{Z}_{\geq 0}.$$

The mixed integrals themselves are classical: they occur as $T_{n+m,n}$ in Amdeberhan et al. [7, Section 5]. The pinned manuscript proves a fixed- m expansion, its late coefficients, and a sharp remainder at optimal truncation. It explicitly leaves the regime in which both exponents increase as a separate question. Here we resolve a transition inside that joint regime.

For fixed m , the first endpoint term is

$$M_{n,m} \sim \frac{\gamma^m n!}{(m+1)^{n+1}}.$$

When m increases, its first neglected local perturbation has size $m \exp[-n/(m+1)]$. Thus it can survive in the limit even though the relevant endpoint x tends to zero. The transition occurs around $n/m = \log m$. The theorem below gives its complete expansion, with explicit first and second corrections. In particular, it does not substitute a growing m into a fixed- m error estimate.

4.1 The transition theorem and its first two corrections

Define the exact constants

$$c = \frac{\zeta(2)}{2\gamma}, \quad d = \frac{\zeta(3)}{3\gamma} - \frac{c^2}{2}, \quad e = \frac{\zeta(4)}{4\gamma} - \frac{c\zeta(3)}{3\gamma} + \frac{c^3}{3}, \quad (4.1)$$

$$\delta = c - \gamma, \quad \varepsilon = \gamma^2 + d, \quad \eta = e + \frac{\gamma^3}{2} - \frac{\gamma^2\delta}{2} + \frac{\gamma\delta^2}{2} - 5\gamma\varepsilon. \quad (4.2)$$

In particular,

$$\delta = 0.84767122475766887678\dots, \quad \varepsilon = 0.01219631601207367653\dots$$

The definitions, rather than these diagnostic decimals, specify the constants in all statements. Notice that $\delta > 0$; for example, $\gamma < 0.6$ and $\zeta(2) > 1.6$ already suffice.

Theorem 4.1 (Uniform joint-exponent transition). *Let*

$$r = m + 1, \quad t = \frac{n + 1}{m + 1}, \quad \lambda = me^{-t}, \quad \mathcal{R}_{n,m} = \frac{(m + 1)^{n+1} M_{n,m}}{\gamma^m n!}.$$

Fix $0 < a < b < \infty$. As $m \rightarrow \infty$, uniformly over positive integers n for which $a \leq \lambda \leq b$,

$$\mathcal{R}_{n,m} = e^{\delta\lambda} \left\{ 1 + \frac{P_1(t, \lambda)}{m} + \frac{P_2(t, \lambda)}{m^2} + O_{a,b} \left(\frac{t^3}{m^3} \right) \right\}, \quad (4.3)$$

where

$$P_1(t, \lambda) = \frac{t}{2} \delta \lambda (1 + \delta \lambda) - 2\gamma \lambda + \varepsilon \lambda^2, \quad (4.4)$$

and

$$\begin{aligned}
P_2(t, \lambda) = & \lambda \left\{ \frac{\delta t^2}{8} - \left(\gamma + \frac{5\delta}{6} \right) t \right\} \\
& + \lambda^2 \left\{ \frac{7\delta^2 t^2}{8} + \left(-3\gamma\delta - \frac{3\delta^2}{2} + 2\varepsilon \right) t + \frac{15\gamma^2}{2} + 3\gamma\delta \right\} \\
& + \lambda^3 \left\{ \frac{3\delta^3 t^2}{4} + \left(-\gamma\delta^2 - \frac{\delta^3}{3} + \frac{5\delta\varepsilon}{2} \right) t + \eta \right\} \\
& + \lambda^4 \left\{ \frac{\delta^4 t^2}{8} + \frac{\delta^2 \varepsilon t}{2} + \frac{\varepsilon^2}{2} \right\}.
\end{aligned} \tag{4.5}$$

More generally, for every fixed integer $J \geq 0$, there are explicitly computable polynomials $P_0 = 1, P_1, \dots, P_J$ such that

$$\mathcal{R}_{n,m} = e^{\delta\lambda} \left\{ \sum_{\ell=0}^J \frac{P_\ell(t, \lambda)}{m^\ell} + O_{a,b,J} \left(\frac{t^{J+1}}{m^{J+1}} \right) \right\}. \tag{4.6}$$

They have degree at most ℓ in t and at most 2ℓ in λ , and

$$P_\ell \in \mathbb{Q}[\gamma, \gamma^{-1}, \zeta(2), \dots, \zeta(\ell+2)][t, \lambda].$$

The finite coefficient prescription in (4.18) below determines every P_ℓ .

Here $t = \log m - \log \lambda = \log m + O_{a,b}(1)$, so the displayed remainders are successive powers of $(\log m)/m$.

Corollary 4.2 (The surviving correction to the endpoint formula). *If $m \rightarrow \infty$ and*

$$\frac{n+1}{m+1} - \log m \longrightarrow s \in \mathbb{R},$$

then

$$\frac{(m+1)^{n+1} M_{n,m}}{\gamma^m n!} \longrightarrow \exp \left[\left(\frac{\zeta(2)}{2\gamma} - \gamma \right) e^{-s} \right]. \tag{4.7}$$

In particular the fixed- m leading term does not have uniform relative error $o(1)$ on this window. At $s = 0$ its missing factor tends to $e^\delta = 2.3342\dots$

Proof. Use $\lambda \rightarrow e^{-s}$ in (4.6) with $J = 0$. The corrections tend to zero because $t/m \rightarrow 0$. $\square$

4.2 Exact gamma averaging and localization at the correct endpoint

Put

$$g(x) = \log \Gamma(1+x), \quad B(x) = \log \Gamma(1-x), \quad C(x) = \frac{B(x)}{\gamma x}, \quad C(0) = 1.$$

The classical convergent log-gamma expansion [6, §5.7] gives

$$\begin{aligned}
g(x) &= -\gamma x + \frac{\zeta(2)}{2} x^2 - \frac{\zeta(3)}{3} x^3 + O(x^4), \\
\log C(x) &= cx + dx^2 + ex^3 + O(x^4).
\end{aligned} \tag{4.8}$$

All these germs are holomorphic in a fixed disk about zero. For $0 < x < 1$, log-convexity of Γ between 1 and 2 implies

$$0 < f(x) \leq -\log x, \quad 0 < B(x) \leq -\log(1-x). \tag{4.9}$$

After $x = e^{-y}$, the normalized integral is the exact expectation

$$\mathcal{R}_{n,m} = \mathbb{E}e^{S(Y)}, \quad S(y) = n \log \left(1 + \frac{g(e^{-y})}{y} \right) + m \log C(e^{-y}), \quad (4.10)$$

where Y has a gamma distribution with shape $q = n + 1 = rt$ and rate $r = m + 1$. Thus

$$\mathbb{E}Y = t, \quad \text{Var } Y = \frac{t}{r}.$$

The powers and logarithms in (4.10) are real, since the original factors are positive.

Lemma 4.3 (Uniform localization). *Under the compact- λ hypotheses of Theorem 4.1, there is $\kappa > 0$ such that*

$$\mathbb{E} \left[e^{S(Y)} \mathbf{1}_{|Y-t|>1} \right] = O_{a,b} \left(e^{-\kappa m/t} \right). \quad (4.11)$$

The same estimate holds with $e^{S(Y)}$ replaced by any fixed power of $|Y - t|$. In particular, the endpoint $x \rightarrow 1$, corresponding to $y \rightarrow 0$, is negligible on the normalization used in the theorem.

Proof. The first summand of $S(y)$ is nonpositive by (4.9). Analyticity of C and $C(0) = 1$ give a constant K such that

$$e^{S(y)} \leq \exp(Kme^{-y}) \quad (y \geq 1).$$

For $y \geq t + 1$ this weight is uniformly bounded, and the gamma Chernoff bound gives a tail $O(e^{-\kappa m/t})$. For example the gamma rate function is $I(u) = u - 1 - \log u$, and $rtI(1 + 1/t) \geq r/(3t)$ for all sufficiently large t .

For the lower tail, let $p(y)$ denote the gamma density and put $v = t - y$. When $1 \leq y \leq t - 1$, we have

$$\frac{p(t-v)}{p(t)} = \exp\{(rt-1)\log(1-v/t) + rv\} \leq t \exp\left(-\frac{rv^2}{2t}\right).$$

Here $1 \leq v \leq t - 1$, and $me^{-(t-v)} = \lambda e^v$. The maximum of e^v/v^2 on this interval is at one of its endpoints; consequently

$$K\lambda e^v \leq \frac{rv^2}{4t} \quad (1 \leq v \leq t - 1)$$

for all sufficiently large m , uniformly in $a \leq \lambda \leq b$. Indeed the two required bounds reduce to $O(1) \leq r/(4t)$ and $O(m/t^2) \leq r/(4t)$. It follows that the weighted lower-tail integral over $[1, t - 1]$ is at most

$$p(t)t^2 e^{-r/(4t)}.$$

Stirling's inequalities give $p(t) = O(\sqrt{r/t})$; the polynomial factor is absorbed by slightly reducing κ .

It remains to bound $0 < y < 1$. Since $1 - e^{-y} \geq y/e$ there,

$$B(e^{-y}) \leq 1 - \log y, \quad (1 - \log y)^m \leq y^{-m}.$$

Using $f(e^{-y}) \leq y$ in the original normalized integral bounds this part by

$$\frac{r^{n+1}}{\gamma^m n!} \int_0^1 y^n (1 - \log y)^m dy \leq \frac{r^{n+1}}{\gamma^m n! (n - m + 1)}. \quad (4.12)$$

Since $n + 1 = rt$ and $t = \log m + O(1)$, the logarithm of this bound is

$$rt(1 - \log t) + O(m + \log n),$$

which is smaller than $-\kappa m/t$ for large m . This also shows directly why the other endpoint cannot compete. The unweighted gamma estimates, with polynomial factors, follow by the same density estimates or by Chernoff's inequality with a slightly smaller exponent. $\square$

4.3 Derivation of the first two corrections

Write $U = Y - t$. On $|U| \leq 1$, set $x = (\lambda/m)e^{-U}$ and use (4.8). Uniformly there, together with every fixed number of U -derivatives,

$$S(t+U) = S_0(U) + \frac{S_1(U)}{m} + \frac{S_2(U)}{m^2} + O(m^{-3}), \quad (4.13)$$

where

$$\begin{aligned} S_0(U) &= \lambda e^{-U} \left(c - \frac{\gamma t}{t+U} \right), \\ S_1(U) &= -\frac{\gamma \lambda e^{-U}(t-1)}{t+U} + \lambda^2 e^{-2U} \left(d + \frac{\gamma c t}{t+U} - \frac{\gamma^2 t}{2(t+U)^2} \right), \\ S_2(U) &= \lambda^2 e^{-2U}(t-1) \left(\frac{\gamma c}{t+U} - \frac{\gamma^2}{2(t+U)^2} \right) \\ &\quad + \lambda^3 e^{-3U} \left(e - \frac{\gamma t(d + c^2/2)}{t+U} + \frac{\gamma^2 c t}{(t+U)^2} - \frac{\gamma^3 t}{3(t+U)^3} \right). \end{aligned} \quad (4.14)$$

The remainders are uniform because $t/(t+U)$ is bounded, $x = O(m^{-1})$, and $n = O(mt)$. The displayed amplitudes and all their fixed-order derivatives are uniformly bounded.

The central gamma moments needed for the first correction are

$$\mathbb{E}U = 0, \quad \mathbb{E}U^2 = \frac{t}{r}, \quad \mathbb{E}U^3 = \frac{2t}{r^2}, \quad \mathbb{E}U^4 = \frac{3t^2}{r^2} + \frac{6t}{r^3}.$$

In particular, for a smooth amplitude H whose first four derivatives are uniformly bounded on $[-1, 1]$, Taylor expansion through the cubic term, followed by the localization lemma, gives

$$\mathbb{E}H(U) = H(0) + \frac{t}{2m} H''(0) + O(t^2/m^2).$$

This notation means that H is averaged on the localized interval; any uniformly bounded extension outside is immaterial up to the exponentially small error already bounded. The signed third moment is used: an absolute third-moment estimate would only give an error of order $(t/m)^{3/2}$.

Since $S_0(0) = \delta\lambda$, the first normalized correction equals

$$S_1(0) + \frac{t}{2}(S_0''(0) + S_0'(0)^2).$$

The derivatives are

$$S_0'(0) = \lambda \left(-\delta + \frac{\gamma}{t} \right), \quad S_0''(0) = \lambda \left(\delta - \frac{2\gamma}{t} - \frac{2\gamma}{t^2} \right).$$

Substitution gives (4.4). The terms involving $1/t$ cancel exactly; this is an algebraic cancellation, not an additional large- t approximation.

For an independently reproducible second-order calculation, define

$$a_j = S_0^{(j)}(0) \quad (1 \leq j \leq 4), \quad b_j = S_1^{(j)}(0) \quad (0 \leq j \leq 2).$$

Then the second correction is exactly

$$\begin{aligned} P_2 &= S_2(0) + \frac{b_0^2}{2} - \frac{t}{2}(a_2 + a_1^2) \\ &\quad + \frac{t}{2}\{b_2 + 2a_1 b_1 + (a_2 + a_1^2)b_0\} \\ &\quad + \frac{t}{3}(a_3 + 3a_1 a_2 + a_1^3) \\ &\quad + \frac{t^2}{8}(a_4 + 4a_1 a_3 + 3a_2^2 + 6a_1^2 a_2 + a_1^4). \end{aligned} \quad (4.15)$$

Expanding this finite formula gives (4.5). To justify the remainder, Taylor-expand the leading amplitude through degree five and bound its degree-six remainder. The central moments satisfy

$$\mathbb{E}U^5 = \frac{20t^2}{r^3} + \frac{24t}{r^4}, \quad \mathbb{E}U^6 = \frac{15t^3}{r^3} + \frac{130t^2}{r^4} + \frac{120t}{r^5}.$$

Use also $\mathbb{E}U^2 = t/m - t/m^2 + O(t/m^3)$ and $\mathbb{E}U^3 = 2t/m^2 + O(t/m^3)$. The m^{-1} amplitude needs Taylor terms only through degree three, and the m^{-2} amplitude only through degree one with a quadratic remainder. All omitted terms are $O(t^3/m^3)$. This proves (4.3) with its claimed uniformity.

4.4 Exact offset-residue identities and a finite all-order recipe

For comparison with the fixed- m manuscript, retain its coefficients

$$A_{k,m} = \frac{1}{k} [x^{k-1}] \Gamma(1+x)^k B(x)^m.$$

Set

$$H(x) = g(x) + \log C(x) = \sum_{j \geq 1} h_j x^j, \quad g(x) = \sum_{j \geq 1} g_j x^j.$$

Then

$$h_1 = \delta, \quad h_2 = d + \gamma c, \quad h_3 = e - \gamma(d + c^2/2), \quad g_1 = -\gamma.$$

Proposition 4.4 (Exact polynomial structure at every fixed offset). *For $j, m \geq 0$,*

$$Q_j(m) := \frac{(m+j+1)A_{m+j+1,m}}{\gamma^m} = [x^j] \exp\{mH(x) + (j+1)g(x)\}. \quad (4.16)$$

In particular Q_j is a polynomial in m of degree j , with leading coefficient $\delta^j/j!$. Its first two terms, for $j \geq 2$, are

$$Q_j(m) = \frac{\delta^j}{j!} m^j + \left\{ \frac{h_2 \delta^{j-2}}{(j-2)!} - \frac{\gamma(j+1)\delta^{j-1}}{(j-1)!} \right\} m^{j-1} + O_j(m^{j-2}). \quad (4.17)$$

For $j = 0, 1$ the identity gives respectively $Q_0 = 1$ and $Q_1 = m\delta - 2\gamma$.

Proof. Write $B(x)^m = (\gamma x)^m C(x)^m$ in the coefficient formula and use $m+j+1 = m + (j+1)$. This gives (4.16) exactly. Each factor of m in the exponential consumes at least one degree of x , proving the degree assertion. The degree- j term uses only $h_1 x$. The degree- $(j-1)$ term either uses $h_2 x^2$ once or $(j+1)g_1 x$ once. These are precisely (4.17). $\square$

The same identity provides a finite prescription for every correction in Theorem 4.1. Let z, w be formal variables and put, for $j \geq 0$,

$$\Theta_j(z, t) = t \sum_{r \geq 1} \frac{(-1)^{r+1} j^{r+1}}{r+1} \left(\frac{z}{1+z} \right)^r,$$

$$K_j(z, w) = \exp \left\{ \delta w + \sum_{r \geq 1} z^r (h_{r+1} w^{r+1} + (j+1)g_r w^r) \right\}.$$

Then

$$\boxed{P_\ell(t, \lambda) = e^{-\delta \lambda} [z^\ell] \sum_{j \geq 0} \lambda^j e^{\Theta_j(z, t)} [w^j] K_j(z, w).} \quad (4.18)$$

Coefficient extraction in z is performed before summing the resulting entire series in λ . This is essential: no convergence of the original infinite pole expansion is asserted.

Why the prescription is finite. At order z^ℓ , only $h_1, \dots, h_{\ell+1}$ and $g_1, \dots, g_\ell$ can occur. Every summand is a finite linear combination of

$$j^p \frac{\delta^{j-d}}{(j-d)!} \lambda^j,$$

with the expression understood to be zero for $j < d$. Their sums are obtained by applying powers of $D_\lambda = \lambda d/d\lambda$ to $\lambda^d e^{\delta\lambda}$. A term of cost z^r in either exponential has total j -degree plus w -degree at most $2r$. Therefore $\deg_\lambda P_\ell \leq 2\ell$; also $\deg_t P_\ell \leq \ell$. The coefficients lie in the ring stated in the theorem, since $h_{\ell+1}$ uses zeta values only through $\zeta(\ell+2)$.

Corollary 4.5 (Two coefficient identities at every asymptotic order). *Let $T_k(u) = e^{-u}(u d/du)^k e^u$ be the Touchard polynomial, equivalently $T_k(u) = \sum_{j=0}^k \{k\}_j u^j$. Then*

$$[t^\ell] P_\ell(t, \lambda) = \frac{T_{2\ell}(\delta\lambda)}{2^\ell \ell!}, \quad (4.19)$$

$$[\lambda^{2\ell}] P_\ell(t, \lambda) = \frac{1}{\ell!} \left(\varepsilon + \frac{\delta^2 t}{2} \right)^\ell. \quad (4.20)$$

These are exact finite polynomial identities, for every $\ell \geq 0$.

Proof. The highest possible power $t^\ell z^\ell$ in (4.18) comes only from the ℓ th power of $tzj^2/2$. Summing $j^{2\ell} \lambda^j \delta^j / j!$ gives (4.19). For the second assertion, every exponent term of order z^r with $r \geq 2$ has total j -degree plus w -degree at most $r+1 < 2r$. Therefore the degree 2ℓ in λ comes only from the ℓ th power of

$$h_2 w^2 + (j+1)g_1 w + \frac{t}{2} j^2.$$

At the highest degree, the summation rule replaces a monomial $w^d j^p$ by $\delta^p \lambda^{d+p}$. The resulting coefficient is $(h_2 + g_1 \delta + t\delta^2/2)^\ell / \ell!$. Since $h_2 + g_1 \delta = \varepsilon$, this is (4.20). $\square$

Why these are the analytic asymptotic coefficients. The localization proof and (4.13) extend to any fixed order. All local amplitudes have bounded derivatives of every fixed order. The centered gamma cumulants are

$$\kappa_j(U) = (j-1)! \frac{t}{r^{j-1}} \quad (j \geq 2), \quad \kappa_1(U) = 0.$$

Consequently, Taylor expansion through degree $2J+1$, with its $2J+2$ remainder, yields (4.6) initially with coefficients that are $e^{\delta\lambda}$ times polynomials in λ, t, t^{-1} . The moment-cumulant formula shows that each omitted contribution is $O(t^{J+1} m^{-J-1})$. The signed odd moments are again used.

To identify these coefficients without retaining spurious negative powers of t , introduce the auxiliary holomorphic family

$$\mathcal{R}(\xi) = \frac{r^{n+1}}{n!} \int_0^\infty (y + g(\xi e^{-y}))^n C(\xi e^{-y})^m e^{-ry} dy.$$

At $\xi = 1$ this is the original normalized integral. Choose a fixed $a_0 > 0$ small enough that g and C are holomorphic on a neighborhood of $|x| \leq a_0$. For $|\xi| \leq a_0$ the displayed family is holomorphic; the exponents n, m are integers, so zeros of the factors introduce no branch choices.

The local expansion is uniform on that complex ξ -disk, with $\xi\lambda$ replacing λ . Here is the necessary tail check, since positivity cannot be used for complex ξ . For $y \geq 1$,

$$\left| 1 + \frac{g(\xi e^{-y})}{y} \right|^n |C(\xi e^{-y})|^m \leq \exp(Kme^{-y} + Kne^{-y}/y).$$

On $y \geq t/2$ this is bounded by $\exp(K'me^{-y})$, so the previous density argument applies. On $Y_0 \leq y \leq t/2$, choose the fixed Y_0 sufficiently large: the last exponent is then at most $rt/16$, whereas the gamma density has a deficit of at least $rt/8$. On $0 \leq y \leq Y_0$, bound the original integrand's two analytic factors by $L^n K_0^m$, with fixed L, K_0 . After multiplication by $r^{n+1}/n!$, its logarithm is $rt(1 - \log t + \log L) + O(m + \log n)$, again negligible. Thus the all-order remainders are uniform for $|\xi| \leq a_0$. Cauchy's coefficient formula consequently permits extraction of any fixed coefficient of ξ^j from those finite asymptotic expansions.

For such a fixed j , take $n \geq j$, which holds eventually in our regime. The binomial expansion is finite, and the elementary integral

$$\binom{n}{h} \frac{r^{n+1}}{n!} \int_0^\infty y^{n-h} e^{-(r+j)y} dy = \frac{1}{h!} \left(\frac{r}{r+j} \right)^{n+1} (r+j)^h$$

shows that the exact ξ^j coefficient is

$$[\xi^j] \mathcal{R}(\xi) = \left(\frac{r}{r+j} \right)^{n+1} Q_j(m). \quad (4.21)$$

The restriction $n \geq j$ ensures that every power of g that can contribute is present in the finite binomial expansion. These operations concern a fixed Taylor coefficient; no divergent infinite sum is integrated.

Now set $z = 1/m$. The exact identities

$$z^j Q_j(1/z) = [w^j] K_j(z, w), \quad e^{jt} \left(\frac{r}{r+j} \right)^{rt} = e^{\Theta_j(z, t)}$$

show that the expansion of (4.21) is exactly the j th coefficient in (4.18), with $\xi\lambda$ in place of λ . Compare successive powers of m^{-1} . Their coefficient functions are Laurent polynomials in t ; a nonzero such function cannot be absorbed by an extra factor m^{-1} when $t = \log m + O(1)$. Thus uniqueness identifies every fixed ξ^j coefficient at every fixed asymptotic order. The local coefficient functions are entire in $\xi\lambda$: they are an exponential $e^{\delta\xi\lambda}$ times a polynomial in $\xi\lambda$. Equality of all their Taylor coefficients identifies them as entire functions, permitting evaluation at $\xi = 1$ without requiring analytic continuation of the full auxiliary integral to that point. This proves the finite recipe and the cancellation of every negative power of t . It completes the all-order part of the theorem.

An independent derivation of P_1 and P_2 . Let $E = e^{\delta\lambda}$, and write $D = \lambda d/d\lambda$. The first two subleading offset-residue sums are

$$\begin{aligned} T_1 &= h_2 \lambda^2 E + g_1 (D+1) (\lambda E), \\ T_2 &= h_3 \lambda^3 E + \frac{h_2^2}{2} \lambda^4 E + g_2 (D+1) (\lambda^2 E) + g_1 h_2 (D+1) (\lambda^3 E) + \frac{g_1^2}{2} (D+1)^2 (\lambda^2 E). \end{aligned}$$

Formula (4.18) yields

$$\begin{aligned} EP_1 &= T_1 + \frac{t}{2} D^2 E, \\ EP_2 &= T_2 + \frac{t}{2} D^2 T_1 + \left(-\frac{t}{2} D^2 - \frac{t}{3} D^3 + \frac{t^2}{8} D^4 \right) E. \end{aligned}$$

These give exactly (4.4)–(4.5); the accompanying exact symbolic verifier checks agreement with the independent gamma-moment derivative calculation.

4.5 Numerical diagnostics and the scope of the advance

The script `verify_joint_moments.py` evaluates the defining integral after $x = e^{-y}$ and then centers and standardizes y using the gamma mean and variance. It does not evaluate a residue expansion. It compares the leading transition function, one correction, and two corrections at three offsets $s = -1, 0, 1$ and several increasing m . The two integration-window choices are also compared. These high-precision checks are diagnostics, not interval certificates; the localization argument and Taylor remainders are the proof.

For a compact example, define $A_J = e^{\delta\lambda} \sum_{\ell=0}^J P_\ell(t, \lambda)/m^\ell$. The following entries use $n = \text{round}((m+1) \log m - 1)$ and the exact t, λ from the theorem. All displayed decimals are rounded numerical approximations, including the errors.

m	n	$\mathcal{R}_{n,m}$	$ \mathcal{R}_{n,m} - A_1 /\mathcal{R}_{n,m}$	$ \mathcal{R}_{n,m} - A_2 /\mathcal{R}_{n,m}$
100	464	2.3971135731	1.1853×10^{-3}	1.1477×10^{-4}
300	1716	2.3595250117	2.3130×10^{-4}	8.0445×10^{-6}
1000	6914	2.3435725531	3.4368×10^{-5}	4.1120×10^{-7}
3000	24026	2.3382795613	5.5591×10^{-6}	2.5343×10^{-8}

The calculation uses 80 decimal digits. Across all 15 cases in the JSON data, changing the integration window from 25 to 30 standard deviations changed the result by at most 6.3×10^{-63} . That comparison is a reproducibility check, not a rigorous bound for the quadrature error or an additional proof of the theorem.

The regime in this theorem is asymmetric: $n \sim m \log m$, with n the larger exponent. Symmetry supplies the corresponding result when the roles of n and m are exchanged. Section 5 treats the complementary proportional regime $n/m \rightarrow c \in (0, \infty)$, proving a unique nondegenerate interior saddle. Neither compact-parameter theorem by itself provides a uniform sharp-truncation result when the reflected exponent grows on the inverse-gamma least-term scale.

Editorial correction to the research status. The pinned fixed- m statements remain valid. Their warnings against substituting a growing m are necessary, as (4.7) demonstrates quantitatively. After integration of this contribution, the open problem should be narrowed from “the simultaneous large-exponent problem” to uniform bridges beyond compact $m \exp[-(n+1)/(m+1)]$ and compact proportional ratios, using also Theorem 5.3. No contradiction to an existing theorem is claimed.

4.6 Further research questions

1. Prove a uniform continuation of the transition when $\lambda = me^{-t}$ also tends to zero or infinity. The compact- λ proof identifies its expansion but does not give useful relative bounds once the polynomials in λ grow with m .
2. Replace the finite interval certificate in Theorem 5.1 by a direct analytic inequality, and determine the best global lower bound for the logarithmic slope ratio. The proportional-saddle uniqueness question itself is settled in the next section.
3. Build a uniform approximation joining the interior saddle to $x \asymp 1/m$. The present theorem supplies the boundary value that such an approximation must reproduce.
4. Study the large- ℓ behavior of the explicitly computable P_ℓ . Does their growth lead to a useful optimal-truncation law inside this transition, and how does it interact with the already proved inverse-gamma late coefficients at fixed m ?
5. Determine the maximal growth of m for which the manuscript’s half-first-omitted-term law remains uniform. The local amplitude $B(-\tau)^m$ changes the saddle when m grows, so a fixed-amplitude argument is insufficient.

6. Extend the exact polynomial recipe to other analytic endpoint pairs $f(x) = -\log x + g(x)$ and $B(x) = b_1x + O(x^2)$. The leading transition constant is suggested by $g_1 + b_2/b_1$; a general theorem should state explicit global endpoint hypotheses, not only a formal local series.

5 A certified unique saddle for every proportional reflected moment

Put

$$f(x) = \log \Gamma(x), \quad F(x) = -\psi(x), \quad P(x) = \psi_1(x) \quad (0 < x < 1).$$

All three functions are strictly positive. The local log-gamma series used below is classical; the new assertion here is the global monotonicity inequality, with its explicit exact certificate, and its application to the proportional moment regime. The monotonicity proof is *computer assisted*: elementary analysis controls both endpoints, and a finite rational interval certificate controls the compact middle. It uses no numerical sampling, floating-point special functions, or unproved decimal values as a proof step. We prove the reflected-ratio inequality directly; no priority claim is made for this inequality or its symmetric special case.

Ordinary convexity of f does not furnish a shortcut. For example, the positive decreasing convex function $f_*(x) = x^{-2} - 1$ has the analogous ratio $J_*(x) = (2-x)/(1+x)$, which decreases strictly. Moreover $\log \log \Gamma(x)$ is convex sufficiently near zero, so global logarithmic concavity is unavailable here. The argument must use more of the structure of the gamma function.

Theorem 5.1 (Certified global log-gamma slope ratio). *Define*

$$J(x) = \frac{F(x)f(1-x)}{F(1-x)f(x)}.$$

Then J is a strictly increasing real-analytic bijection from $(0,1)$ onto $(0,\infty)$. More quantitatively,

$$\frac{J'(x)}{J(x)} > \frac{176}{225} \quad (0 < x < 1). \quad (5.1)$$

Also $J(1-x) = J(x)^{-1}$, $J(1/2) = 1$, and $J(x) \sim 1/\log(1/x)$ as $x \downarrow 0$.

5.1 Elementary endpoint proof

Differentiating the displayed definition gives

$$\mathcal{H}(x) := \frac{J'(x)}{J(x)} = \frac{F(x)}{f(x)} + \frac{F(1-x)}{f(1-x)} - \frac{P(x)}{F(x)} - \frac{P(1-x)}{F(1-x)}. \quad (5.2)$$

In particular $\mathcal{H}(1-x) = \mathcal{H}(x)$. The familiar convergent expansions [6, §5.7], valid for $|x| < 1$ with the indicated real logarithm for $x > 0$, are

$$\begin{aligned}
f(x) &= -\log x - \gamma x + \sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} x^n, \\
f(1-x) &= \gamma x + \sum_{n \geq 2} \frac{\zeta(n)}{n} x^n, \\
F(x) &= \frac{1}{x} + \gamma + \sum_{n \geq 2} (-1)^{n-1} \zeta(n) x^{n-1}, \\
F(1-x) &= \gamma + \sum_{n \geq 2} \zeta(n) x^{n-1}, \\
P(x) &= \frac{1}{x^2} + \sum_{n \geq 2} (-1)^n (n-1) \zeta(n) x^{n-2}, \\
P(1-x) &= \sum_{n \geq 2} (n-1) \zeta(n) x^{n-2}.
\end{aligned} \tag{5.3}$$

The differentiated series converge uniformly on every smaller closed disk. Only the elementary bounds $1/2 < \gamma < 3/5$ and $\zeta(n) \leq \zeta(2) < 2$ are needed in the endpoint argument; the certificate below also independently encloses these constants.

For $0 < x \leq 1/16$, rewrite (5.2) as

$$\mathcal{H}(x) = \frac{F(x)}{f(x)} - \frac{P(1-x)}{F(1-x)} + \left(\frac{F(1-x)}{f(1-x)} - \frac{1}{x} \right) + \left(\frac{1}{x} - \frac{P(x)}{F(x)} \right).$$

Both parenthesized terms are nonnegative. For the first,

$$xF(1-x) - f(1-x) = \sum_{n \geq 2} \frac{n-1}{n} \zeta(n) x^n > 0.$$

For the second, the convergent series yields

$$\begin{aligned}
F(x) - xP(x) &= \gamma + \sum_{k \geq 1} (-1)^k (k+1) \zeta(k+1) x^k \\
&> \frac{1}{2} - 2((1-x)^{-2} - 1) \geq \frac{1}{2} - \frac{62}{225} = \frac{101}{450} > 0.
\end{aligned}$$

Also

$$F(x) > \frac{1}{x} + \frac{1}{2} - \frac{2x}{1-x} > \frac{1}{x}.$$

Log-convexity of Γ between one and two gives $0 < f(x) \leq -\log x$. Since $x \log(1/x)$ increases on $(0, 1/16]$ and $\log 16 < 3$, it follows that

$$\frac{F(x)}{f(x)} > \frac{1}{x \log(1/x)} > \frac{16}{3}.$$

The inequality $\log 16 < 3$ follows, for example, from the strict trapezoidal upper bound $\log 2 < 3/4$ for the integral of $1/t$ on $[1, 2]$. Finally,

$$\frac{P(1-x)}{F(1-x)} < \frac{4}{(1-x)^2} \leq \frac{1024}{225}.$$

Combining the four bounds proves $\mathcal{H}(x) > 16/3 - 1024/225 = 176/225$ on $(0, 1/16]$. Reflection proves the same bound on $[15/16, 1)$.

5.2 The compact interval certificate

Lemma 5.2 (Exact finite compact certificate). *For every $x \in [1/16, 1/2]$,*

$$\mathcal{H}(x) > \frac{7}{4}.$$

Proof by a finite rational certificate. The dependency-free program `certify_gamma_saddle.py` recomputes the following enclosure procedure from scratch. Every interval endpoint is an integer multiple of 2^{-100} . Addition is exact; multiplication, inversion and division round outward using integer floor and ceiling. Thus all intermediate intervals contain the corresponding exact real quantities, by induction on the arithmetic operations. No stored interval endpoint needs to be trusted by the verifier.

For $2 \leq n \leq 40$, enclose $\zeta(n)$ using the first 64 terms and the integral-test bounds

$$\sum_{k=1}^{64} k^{-n} + \frac{65^{1-n}}{n-1} \leq \zeta(n) \leq \sum_{k=1}^{64} k^{-n} + \frac{64^{1-n}}{n-1}.$$

For γ , use

$$\gamma = \sum_{k \geq 1} \left(\frac{1}{k} - \log(1 + 1/k) \right),$$

and the alternating logarithm bounds. The tail after K terms lies between

$$\frac{1}{2(K+1)} - \frac{1}{6K^2} \quad \text{and} \quad \frac{1}{2K}.$$

Taking $K = 64$ gives a rational interval for $H_{64} - \log 65$ plus that tail. The computed enclosure lies strictly inside $(1/2, 3/5)$. Logarithms of positive rationals are enclosed by range reduction to $1 \leq u \leq 2$ and

$$\log u = 2 \sum_{k=0}^{47} \frac{y^{2k+1}}{2k+1} + R, \quad y = \frac{u-1}{u+1}, \quad 0 \leq R \leq \frac{2y^{97}}{97(1-y^2)}.$$

The same formula supplies $\log 2$ for the range reduction. All arguments and finite sums are rational.

Truncate each series in (5.3) at $N = 40$. On an interval $0 < \ell \leq x \leq u \leq 1/2$, the absolute omitted tails for f, F, P are bounded respectively by

$$T_f = \frac{2u^{N+1}}{(N+1)(1-u)}, \quad T_F = \frac{2u^N}{1-u}, \quad T_P = \frac{2Nu^{N-1}}{(1-u)^2}. \quad (5.4)$$

These follow directly from $\zeta(n) < 2$ and geometric sums. The positive reflected series receive one-sided positive tails; the alternating series receive symmetric tails. Their polynomial parts are evaluated by interval Horner arithmetic. Every denominator appearing in (5.2) is checked to have positive lower bound.

Partition $[1/16, 1/2]$ into the following 256 closed intervals:

$$\left[\frac{1}{16} + \frac{7j}{4096}, \frac{1}{16} + \frac{7(j+1)}{4096} \right], \quad 0 \leq j < 256.$$

For each entire interval, the program evaluates (5.2) with the rigorous enclosures above. The smallest lower endpoint obtained is exactly

$$\frac{2237344070433247164830307425599}{1267650600228229401496703205376} > \frac{7}{4}.$$

It occurs on $[2041/4096, 1/2]$. The program tests the inequality with integer arithmetic; the saved JSON file records every interval's lower numerator for inspection. Thus the finite certificate proves the asserted inequality on the complete compact interval. $\square$

The endpoint proof, Lemma 5.2, and reflection now prove (5.1). The expansions

$$f(x) \sim \log(1/x), \quad F(x) \sim 1/x, \quad f(1-x) \sim \gamma x, \quad F(1-x) \sim \gamma$$

give $J(x) \sim 1/\log(1/x)$. Reciprocal symmetry gives the opposite endpoint limit. Positivity of J' proves strict monotonicity; the endpoint limits prove surjectivity. Real analyticity and $J' > 0$ prove that its inverse is also real analytic. This completes Theorem 5.1.

5.3 The proportional all-orders moment theorem

For real $n, m > 0$, define

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx.$$

These integrals converge. In particular, the following result applies to the integer moment family in the manuscript.

Theorem 5.3 (Unique proportional saddle and uniform expansion). *For $c > 0$, put*

$$\Phi_c(x) = c \log f(x) + \log f(1-x), \quad s(c) = J^{-1}(1/c), \quad A_c = -\Phi_c''(s(c)).$$

Then $s(c)$ is the unique global maximizer of Φ_c , and

$$A_c = \frac{F(1-s(c))}{f(1-s(c))} \mathcal{H}(s(c)) > 0. \quad (5.5)$$

For every compact $C \subset (0, \infty)$ and every fixed $N \geq 0$, uniformly for $c = n/m \in C$ as $m \rightarrow \infty$,

$$M_{n,m} = f(s(c))^n f(1-s(c))^m \sqrt{\frac{2\pi}{mA_c}} \left\{ \sum_{j=0}^N \frac{L_j(c)}{m^j} + O_{C,N}(m^{-N-1}) \right\}. \quad (5.6)$$

The coefficient functions are real analytic on $(0, \infty)$, with $L_0 = 1$ and

$$L_1(c) = \frac{\Phi_c^{(4)}(s(c))}{8A_c^2} + \frac{5\Phi_c^{(3)}(s(c))^2}{24A_c^3}. \quad (5.7)$$

Every higher coefficient is given by the finite Gaussian prescription below.

Proof. Let $g = F/f$. The derivative has the decisive factorization

$$\Phi_c'(x) = g(1-x)(1-cJ(x)).$$

Theorem 5.1 gives exactly one zero, with positive derivative sign before the zero and negative sign after it. Differentiating at that zero gives

$$\Phi_c''(s) = -cg(1-s)J'(s) = -g(1-s)\mathcal{H}(s),$$

which proves (5.5) and nondegeneracy.

For completeness, the uniform endpoint control needed by Laplace's method follows from elementary bounds. At $0 < x \leq 1/2$, $f(x) \leq -\log x$, and (5.3) gives

$$f(1-x) \leq \gamma x + \frac{\zeta(2)x^2}{2(1-x)} < 2x.$$

Thus as $x \downarrow 0$,

$$\Phi_c(x) \leq c \log \log(1/x) + \log(2x) \longrightarrow -\infty$$

uniformly for c in any fixed compact positive interval. Reflection gives the same statement at $x \uparrow 1$, using the positive lower bound on c . Consequently all saddles stay inside a fixed compact subinterval. By uniqueness and compactness, outside any sufficiently small fixed neighborhood of the saddle there is a uniform positive gap below $\Phi_c(s(c))$. Integration of that bound over an interval of length one makes the complementary contribution exponentially small relative to the saddle term.

On a common neighborhood of the saddle, every fixed phase derivative is bounded uniformly in c , and A_c is uniformly bounded below by a positive number. Taylor-expand the analytic phase and set $x = s(c) + u/\sqrt{m}$. A uniform local Gaussian bound follows from the strictly negative second derivative and compactness. Taylor expansion to any fixed order, with its uniformly bounded remainder, followed by Gaussian integration proves (5.6). Explicitly, $L_j(c)$ is the coefficient of t^{2j} in the normalized Gaussian expectation, for $Z \sim N(0, A_c^{-1})$, of

$$\exp \left\{ \sum_{k \geq 3} \frac{\Phi_c^{(k)}(s(c))}{k!} Z^k t^{k-2} \right\}.$$

Only finitely many derivatives occur at every order. The t^2 coefficient uses the fourth Gaussian moment and half the square of the cubic term, giving (5.7). The same finite formula proves real analyticity of the coefficient functions. $\square$

Corollary 5.4 (Balanced moments). *Put*

$$f_0 = \frac{1}{2} \log \pi, \quad F_0 = \gamma + 2 \log 2, \quad A_{\text{bal}} = \frac{2F_0^2}{f_0^2} - \frac{\pi^2}{f_0} > 0.$$

Then the unique saddle of $M_{m,m}$ is $1/2$, and

$$M_{m,m} \sim f_0^{2m} \sqrt{\frac{2\pi}{mA_{\text{bal}}}}.$$

It has the full expansion (5.6); in its first correction the cubic phase derivative vanishes by reflection symmetry.

Proof. Use $J(1/2) = 1$, $\psi(1/2) = -\gamma - 2 \log 2$, and $\psi_1(1/2) = \pi^2/2$ in (5.5). $\square$

Corollary 5.5 (Saddle motion and a strictly convex limiting rate). *Define $\mathcal{V}(c) = \Phi_c(s(c))$. Then*

$$s(1/c) = 1 - s(c), \quad s'(c) = -\frac{g(s(c))}{A_c} < 0,$$

and

$$\mathcal{V}'(c) = \log f(s(c)), \quad \mathcal{V}''(c) = \frac{g(s(c))^2}{A_c} > 0, \quad \mathcal{V}(c) = c\mathcal{V}(1/c).$$

In particular $m^{-1} \log M_{cm,m} \rightarrow \mathcal{V}(c)$ uniformly on compact positive c -intervals.

Proof. Reciprocal symmetry of J gives the first identity. Differentiate the stationary equation $\Phi'_c(s(c)) = 0$ for s' , and differentiate the phase value for the two envelope identities. The reciprocity of $\mathcal{V}$ follows by substituting $1 - s(c)$ in $\Phi_{1/c}$. The last assertion follows from (5.6). $\square$

What remains open after this theorem. The proportional regime on compact positive exponent ratios is now resolved, including uniqueness, curvature and all fixed asymptotic orders. A proof of the global inequality (5.1) using only an analytic inequality, without the finite interval certificate, remains a worthwhile simplification problem. The uniform bridge joining this compact- c regime to the logarithmic transition of Theorem 4.1, and extending beyond that window, remains a separate problem. Compact- c uniformity alone does not justify substitution of a growing exponent ratio.

6 Integral distribution coordinates and primitive-grid torsion

The complex character coordinates of the manuscript diagonalize the weighted distribution relations, including their exceptional weights. They do not by themselves determine the integral lattice or justify reduction of that diagonalization modulo a prime dividing a character group order. We give a concrete integral model and then compute a primitive-grid cokernel completely at a product of two distinct primes. The latter calculation reveals torsion invisible to complex rank and jet computations.

The general freeness and resolution method belongs to the established theory of universal distributions, originating with Kubert's ordinary distribution theorem [8]. In particular, Ouyang's universal norm distribution has a free-basis theorem and an Anderson resolution [10, Proposition 4.2 and Theorem 5.1]; the ordinary case is treated in [9]. The purpose of the first part below is to supply an explicit, finite, all-divisor grid version with arbitrary weights, arbitrary coefficient rings, and actual grid points as basis vectors. The two-prime cokernel, its integral Smith factors, and its sharp denominator consequence are the specific additional calculations obtained here. No global priority assertion is made for these formulas.

6.1 A basis which survives every specialization

Let $q = \prod_{p|q} p^{e_p} \geq 2$, let R be any commutative unital ring, and choose $t_p \in R$. Put $G_q = (1/q)\mathbb{Z}/\mathbb{Z}$, $V_q = R[G_q]$, and

$$D_p(x) = \sum_{\substack{y \in G_q \\ py=x}} [y] - t_p[x], \quad x \in G_{q/p}, \quad U_q(R, \mathbf{t}) = V_q / \langle D_p(x) \rangle.$$

Thus $G_{q/p} = pG_q$; no distribution row outside its legitimate source is used. All tensor products in this section are over R .

For $m = p^e$, define the following set of local residues:

$$B(p^e) = \{0\} \cup \{p^{e-1} + 1, p^{e-1} + 2, \dots, p^e - 1\}. \quad (6.1)$$

An empty upper interval is allowed; in particular $B(2) = \{0\}$. Use the additive primary decomposition of G_q to put

$$\mathcal{B}_q = \left\{ \left[\sum_{p|q} \frac{b_p}{p^{e_p}} \right] : b_p \in B(p^{e_p}) \right\}. \quad (6.2)$$

Distinct choices give distinct grid points.

Theorem 6.1 (Integral basis and arbitrary base change). *For every R and every choice of the weights, the images of $\mathcal{B}_q$ form an R -basis of $U_q(R, \mathbf{t})$. Consequently*

$$U_q(R, \mathbf{t}) \cong R^{\varphi(q)}.$$

This basis is independent of the weights and contains $[0]$. In particular, imposing the homogeneous endpoint anchor $[0] = 0$ leaves a free module of rank $\varphi(q) - 1$ over every R .

Over the universal ring $\mathbb{Z}[t_p : p | q]$, every grid point has unique integral polynomial coordinates in this basis. For a point $x = \sum_p x_p$ whose p -primary component has additive order p^{j_p} , each coordinate polynomial has degree at most j_p in t_p .

For Hurwitz evaluation, use the representative of a grid point in $(0, 1]$, with $[0]$ evaluated as $\zeta(s, 1) = \zeta(s)$. The term “integral” refers to formal coordinates. It asserts neither linear independence of evaluated zeta or polylogarithm values nor an integral-basis assertion for a field generated by such values.

6.1.1 The finite resolution and its filtration

Order the prime divisors of q . For $S \subseteq \{p : p \mid q\}$ set $d_S = \prod_{p \in S} p$ and

$$C_r = \bigoplus_{|S|=r} R[G_{q/d_S}].$$

Write a generator of its S -summand as $[x, S]$. For $p \in S$, define

$$\partial_p[x, S] = \sum_{\substack{y \in G_{q/d_{S \setminus \{p\}}} \\ py=x}} [y, S \setminus \{p\}] - t_p[x, S \setminus \{p\}],$$

and let ∂ be their sum with the usual exterior-algebra signs. The transfers and inclusions for distinct primes commute. For example, composing two transfers enumerates exactly the $p\ell$ preimages of x , while a transfer by p cannot raise the ℓ -primary order when $p \neq \ell$. Thus $\partial^2 = 0$ and $H_0(C_\bullet) = U_q(R, \mathfrak{t})$.

For a p -primary point x_p , let $h_p(x_p)$ be its additive-order exponent, with $h_p(0) = 0$. Give $[x, S]$ filtration degree

$$|S| + \sum_{p \mid q} h_p(x_p). \quad (6.3)$$

The degree lies between zero and $\sum_p e_p$. In a p -transfer, every preimage other than the zero preimage of $x_p = 0$ raises h_p by one; all the other primary heights are unchanged. The leading part of ∂ preserves (6.3). The inclusion term and the zero preimage lower it by one.

There is a minor coordinate twist which must be accounted for: a p -preimage multiplies every other primary component by p^{-1} . In the S -summand replace its ℓ -primary coordinate by

$$x_\ell \longmapsto \left(\prod_{p \in S, p \neq \ell} p \right)^{-1} x_\ell.$$

All these inverses are units on the relevant primary cyclic group, so this is merely a permutation of the grid basis. It preserves heights. In the new coordinates, the leading p -transfer acts only on the p -primary factor. In degree zero, where S is empty, this coordinate change is the identity.

The associated graded complex is therefore the tensor product, over $p \mid q$, of the two-term complexes

$$R[G_{p^{e_p-1}}] \xrightarrow{N_p^+} R[G_{p^{e_p}}], \quad N_p^+[x] = \sum_{\substack{py=x \\ h_p(y)=h_p(x)+1}} [y]. \quad (6.4)$$

The local domain is in homological degree one, with its height shifted by one. The supports of distinct $N_p^+[x]$ are disjoint. Each support contains exactly one residue in $\{1, \dots, p^{e_p-1}\}$; this gives a coefficient-one pivot. Hence N_p^+ is split injective, and its free cokernel has the actual local grid basis (6.1).

A tensor product of these split two-term complexes has zero positive homology and degree-zero homology freely generated by $\mathcal{B}_q$. This can be seen without a flatness hypothesis: split each local target into the image of N_p^+ and its chosen free complement. The tensor product becomes a direct sum of its free degree-zero complement and contractible complexes.

For completeness, finite-filtration lifting gives the requisite conclusion over any ring. If z is a positive-degree cycle, its leading filtration term is a boundary in the associated graded complex. Lift a preimage and subtract its actual boundary from z . This lowers the filtration, so induction makes z an actual boundary. Similarly, if $\partial y \in C_0$ has lower filtration than $y \in C_1$, the leading term of y is a degree-one cycle for the leading differential. Subtract the boundary of a degree-two lift, leaving ∂y unchanged and lowering the filtration of y . Repetition proves that the image of $\partial : C_1 \rightarrow C_0$ is strict for the filtration. Therefore the associated graded of U_q is precisely the

degree-zero homology computed above. Its specified basis lifts to an actual basis: leading terms prove independence, and descending filtration proves spanning. There are $\prod_p(p^{e_p} - p^{e_p-1}) = \varphi(q)$ basis points. This proves the basis assertion and also proves the following resolution statement.

Corollary 6.2 (All relation syzygies). *The augmented finite complex*

$$0 \longrightarrow C_{\omega(q)} \longrightarrow \cdots \longrightarrow C_1 \longrightarrow C_0 \longrightarrow U_q(R, \mathbf{t}) \longrightarrow 0$$

is exact. It is split exact as a complex of R -modules. In particular, every relation among prime-distribution rows is generated by the commuting-prime two-step relations, and all higher relations are accounted for by the same complex.

To justify the last splitting assertion, U_q is free, so $C_0 \rightarrow U_q$ splits. Its kernel is projective; the surjection from C_1 onto that kernel then splits. Continue inductively. No claim about semisimplicity or character projectors is involved.

6.1.2 A terminating polynomial reduction algorithm

Represent a point by a tuple $x = (a_p \bmod p^{e_p})_p$. If every local residue belongs to $B(p^{e_p})$, it is already a basis point. Otherwise choose any local pivot $1 \leq a_p \leq p^{e_p-1}$. All the other p -preimages of px have local p -residues outside this pivot interval. The distribution row rewrites

$$[x] = t_p[px] - \sum_{j=1}^{p-1} \left[x + \frac{j}{p} \right], \quad (6.5)$$

where every point is taken in G_q and the p -primary component of x is the chosen pivot. The right-hand siblings may include the point with zero p -component. That point has lower primary height.

The nonzero siblings keep the total primary height and remove one pivot. The term px lowers the p -primary height by exactly one; multiplication by p preserves all the other primary heights. Consequently the lexicographically ordered pair

$$(\text{total primary height, number of local pivots})$$

strictly decreases in every recursive branch. Reduction terminates, and uniqueness follows from Theorem 6.1. Every occurrence of t_p lowers the p -height by one. This proves the degree bound asserted in that theorem. A memoized implementation only needs the q grid states and performs exact additions and monomial shifts.

Example 6.3 (A uniform four-point coordinate grid at level twelve). With the primes ordered as 2, 3, the basis is

$$[0], \quad [2/3], \quad [3/4], \quad [5/12].$$

Write $a = t_2$, $b = t_3$. For example, reduction gives

$$[1/12] = a(a-1)b[0] - (b+1)[3/4] - [5/12], \quad (6.6)$$

$$[1/6] = a(b-1)[0] - (a+1)[2/3], \quad (6.7)$$

$$[7/12] = a(b-a)[0] - a(a+1)[2/3] + (b+1)[3/4] + [5/12], \quad (6.8)$$

$$[11/12] = -a(b-1)[0] + a(a+1)[2/3] - [5/12]. \quad (6.9)$$

These are identities over $\mathbb{Z}[a, b]$, including $a = b = 1$ and characteristic two. Analytic evaluation substitutes $t_p = p^s$ for Hurwitz zeta and $t_p = p^{1-s}$ for cyclotomic polylogarithms. At a spectral pole, evaluate the full meromorphic identity before taking a limit; a vanishing coefficient times $\zeta(s)$ can have a nonzero finite contribution.

6.2 The integral meaning of the primitive determinant

Let $W_q = R^{U(q)}$ be the formal span of the primitive q -grid points, and let $P_q : W_q \rightarrow U_q$ be their coordinate map in the integral basis above. This coordinate map need not be injective after a singular specialization. Over $\mathbb{Z}[t_p]$, it is a square matrix of integral polynomials. The complex Fourier/conductor calculation in the manuscript gives

$$\Delta_q = \prod_{p^e \parallel q} t_p^{\varphi(q/p^e)(p^{e-1}-1)} (t_p^{h_p} - 1)^{\varphi(q/p^e)/h_p}, \quad h_p = \text{ord}_{q/p^e}(p),$$

where the order modulo one is one. We now remove the ambiguity from its choice of complex bases.

Theorem 6.4 (Integral primitive determinant). *With any fixed orderings of the integral and primitive grid bases,*

$$\det P_q = \pm \Delta_q. \quad (6.10)$$

Consequently, for integer weights with $\Delta_q \neq 0$, the integral index of the primitive-grid span is exactly $|\Delta_q|$. For an arbitrary coefficient ring the primitive grid is a basis if and only if Δ_q is a unit.

Proof. Over $\mathbb{C}[t_p]$, both the integral basis established above and the conductor basis of the manuscript are bases of the same free module. Their transition determinant is a nonzero constant, since the only units of this polynomial ring are nonzero constants. The primitive Fourier transform is also constant. The manuscript's diagonal determinant therefore implies $\det P_q = c\Delta_q$ for a nonzero constant c . The highest monomial of Δ_q has coefficient one, whereas $\det P_q$ has integer coefficients. Hence $c \in \mathbb{Z}$.

It remains to show that no prime divides c . Over any field K , regard the weights as algebraically independent. Put $z_p = t_p^{-1}$ and, for $u \in U(q)$, form the formal multivariate series

$$F_{r,u}(\mathbf{z}) = \sum_{\substack{(n_p) \in \mathbb{Z}_{\geq 0}^{\omega(q)} \\ u \prod_p p^{n_p} \equiv r \pmod{q}}} \prod_p z_p^{n_p}.$$

The primitive rows form the identity matrix. If $p \mid r$, every term has $n_p \geq 1$, so removing one factor of p proves exactly the p -distribution equation. These are rational functions: on the finite residue grid, the pushforward M_p satisfies $M_p^{e_p+h_p} = M_p^{e_p}$, and summing its powers gives a rational resolvent with denominator $1 - z_p^{h_p}$. Thus this construction is valid over $K(t_p)$ in every characteristic and gives an extension of arbitrary primitive-grid data. Since the full quotient has dimension $\varphi(q)$ there, P_q is invertible over $K(t_p)$.

If a prime divided c , reducing $\det P_q = c\Delta_q$ modulo that prime would give the zero polynomial, contradicting this generic invertibility. Therefore $c = \pm 1$. The remaining assertions follow from the determinant criterion for a square matrix over a ring and, over $\mathbb{Z}$, the index formula for a full-rank sublattice. $\square$

6.3 Complete two-prime primitive cokernels

Assume now that $q = p\ell$ for distinct primes. Write

$$\begin{aligned} a &= t_p, & b &= t_\ell, & h_p &= \text{ord}_p(\ell), & h_\ell &= \text{ord}_\ell(p), \\ c_p &= (p-1)/h_p, & c_\ell &= (\ell-1)/h_\ell, \\ S_p &= 1 + b + \cdots + b^{h_p-1}, & S_\ell &= 1 + a + \cdots + a^{h_\ell-1}, \\ D_p &= b^{h_p} - 1 = (b-1)S_p, & D_\ell &= a^{h_\ell} - 1 = (a-1)S_\ell. \end{aligned} \quad (6.11)$$

Let $T_q = \text{coker } P_q$.

Theorem 6.5 (Two-prime cokernel over an arbitrary ring). *There is an R -module isomorphism*

$$T_{p\ell} \cong (R/(D_p))^{c_p-1} \oplus (R/(D_\ell))^{c_\ell-1} \oplus C(a, b), \quad (6.12)$$

where the three-generator module $C(a, b)$ has generators e, X, Y and relations

$$(a-1)(b-1)e = 0, \quad S_p X = (a-1)e, \quad S_\ell Y = (b-1)e. \quad (6.13)$$

No weight, geometric sum, or integer group order is inverted.

Proof. Kill the primitive $p\ell$ -grid symbols in the original presentation. The remaining generators are $e = [0]$, the $p-1$ points of exact denominator p , and the $\ell-1$ points of exact denominator ℓ . The source of a p -distribution row has denominator either one or ℓ , and the source of an ℓ -row has denominator either one or p . Thus every relation is one of

$$\begin{aligned} \sum_{u \in U(p)} [u/p] &= (a-1)e, & \sum_{v \in U(\ell)} [v/\ell] &= (b-1)e, \\ [\ell^{-1}u/p] &= b[u/p], & [p^{-1}v/\ell] &= a[v/\ell]. \end{aligned}$$

In the last two rows the inverses are residue-class inverses.

Multiplication by ℓ^{-1} on $U(p)$ has c_p cycles, each of length h_p . Along one cycle the relations successively eliminate all but a single generator X_i , with the closing relation $D_p X_i = 0$. Its contribution to the first norm row is $S_p X_i$. The other layer similarly gives generators Y_j , relations $D_\ell Y_j = 0$, and norm contribution $S_\ell Y_j$.

The integer coordinate change from $X_1, \dots, X_{c_p}$ to $X = \sum_i X_i$ and any $c_p - 1$ of the original generators is unimodular. Since every cycle has the same annihilator, the $c_p - 1$ other coordinates split off as $R/(D_p)$. Apply the same operation to the Y -cycles. The remaining presentation has the two norm relations in (6.13), together with $D_p X = 0$ and $D_\ell Y = 0$. Modulo the norm relations, these last two are both equivalent to $(a-1)(b-1)e = 0$, because

$$D_p X = (b-1)S_p X = (a-1)(b-1)e, \quad D_\ell Y = (a-1)S_\ell Y = (a-1)(b-1)e.$$

The replacement works in both directions and uses no cancellation. This proves the asserted presentation over arbitrary R . $\square$

6.3.1 The exact ordinary-weight torsion and modular ranks

Corollary 6.6 (Ordinary-weight primitive torsion). *For $R = \mathbb{Z}$ and $a = b = 1$,*

$$T_{p\ell} \cong \mathbb{Z}^{c_p+c_\ell-1} \oplus \mathbb{Z}/h_p\mathbb{Z} \oplus \mathbb{Z}/h_\ell\mathbb{Z}. \quad (6.14)$$

A cyclic factor of order one means the zero module. For a rational prime r , the primitive-grid inclusion over $\mathbb{F}_r$ at these weights has rank

$$\varphi(p\ell) - (c_p + c_\ell - 1) - \mathbf{1}_{r|h_p} - \mathbf{1}_{r|h_\ell}. \quad (6.15)$$

Proof. At $a = b = 1$ the direct cyclic factors in (6.12) are free copies of $\mathbb{Z}$, and the core relations are simply $h_p X = 0$, $h_\ell Y = 0$, with e free. This gives (6.14). Tensor its presentation with $\mathbb{F}_r$. Each free coordinate contributes one dimension to the cokernel, and a finite cyclic coordinate contributes one exactly when r divides its order. Subtract from the full quotient dimension $\varphi(p\ell)$ furnished by Theorem 6.1. $\square$

For example, $q = 15$ has $h_3 = 2$, $h_5 = 4$, and $c_3 = c_5 = 1$. Its ordinary primitive cokernel is $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$. The primitive rank is seven in characteristic zero and five in characteristic two. At $q = 21$, the corresponding cokernel is $\mathbb{Z}^2 \oplus \mathbb{Z}/6\mathbb{Z}$, giving ranks ten in characteristic zero and nine in characteristics two and three. For $q = 35$, it is $\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z}$. These are failures of a particular integral coordinate inclusion; the full weighted distribution quotient remains free in every case.

6.3.2 Smith factors at nonexceptional integer weights

Corollary 6.7 (Explicit Smith factors for the core). *Let $a, b \in \mathbb{Z}$, and assume $D_p D_\ell \neq 0$. Define positive integers*

$$g_1 = \gcd(a-1, b-1, S_p, S_\ell), \quad g_2 = \gcd(D_p, D_\ell, S_p S_\ell).$$

Then

$$C(a, b) \cong \mathbb{Z}/g_1\mathbb{Z} \oplus \mathbb{Z}/(g_2/g_1)\mathbb{Z} \oplus \mathbb{Z}/(|D_p D_\ell|/g_2)\mathbb{Z}. \quad (6.16)$$

Together with (6.12), this specifies the entire finite primitive-grid cokernel as a direct sum of cyclic groups. Its order is $|D_p|^{c_p} |D_\ell|^{c_\ell}$ and its exponent is

$$\boxed{E_{p\ell}(a, b) = \text{lcm}(|D_p|, |D_\ell|, |(a-1)(b-1)|)}. \quad (6.17)$$

Proof. A row-relation matrix for the core is

$$\begin{pmatrix} (a-1)(b-1) & 0 & 0 \\ -(a-1) & S_p & 0 \\ -(b-1) & 0 & S_\ell \end{pmatrix}.$$

The greatest common divisor of its entries is g_1 . The gcd of its 2×2 minors is g_2 : the three potentially necessary minors are D_ℓ , D_p , and $S_p S_\ell$, and every other minor is zero or a multiple of one of these. Its determinant is $D_p D_\ell$. The determinantal-divisor characterization of integer Smith form therefore gives (6.16). The last invariant factor is a multiple of both D_p and D_ℓ , since g_2 divides both, so the extra cyclic summands in (6.12) do not change the exponent. Finally,

$$\frac{|D_p D_\ell|}{\gcd(D_p, D_\ell, S_p S_\ell)} = \text{lcm}(|D_p|, |D_\ell|, |(a-1)(b-1)|),$$

by taking prime valuations, or by using $D_p D_\ell = (a-1)(b-1)S_p S_\ell$. $\square$

6.4 Sharp rational reduction denominators

Suppose the integer weights are nonexceptional and express every grid point in the primitive basis over $\mathbb{Q}$. Let E be the least positive integer clearing all entries of the resulting extension matrix. Because $\mathcal{B}_q$ itself consists of actual grid points, this is also the least positive integer for which EP_q^{-1} is integral. Equivalently it is the exponent of coker P_q .

Corollary 6.8 (Exact two-prime denominator). *At $q = p\ell$ and integer weights a, b with $D_p D_\ell \neq 0$, the least common denominator of all primitive-grid extension coefficients is precisely (6.17). In particular, for Hurwitz weights $a = p^k$, $b = \ell^k$, $k \geq 1$, it is*

$$\text{lcm}\left(\ell^{k \text{ord}_p(\ell)} - 1, p^{k \text{ord}_\ell(p)} - 1, (p^k - 1)(\ell^k - 1)\right). \quad (6.18)$$

The sharpness also follows directly from the manuscript's finite residue formula. Points of exact denominator p have nonzero coefficients with reduced denominator D_p , points of exact denominator ℓ give D_ℓ , and the endpoint coefficient is $1/((a-1)(b-1))$. The relevant one-prime numerators are powers of the corresponding weight, which are coprime to its power minus one. Thus each of the three integers in the least common multiple is required. The Smith calculation determines substantially more than this denominator alone.

At $q = 15$, $k = 1$, the exact denominator is $\text{lcm}(24, 80, 8) = 240$, whereas the general product bound in the manuscript is $24 \cdot 80 = 1920$. The full primitive index is 1920 and its nontrivial Smith factors are 2, 4, 240. At $q = 6$, $k = 5$, the exact denominator is $\text{lcm}(242, 1023, 7502) = 22506$, whereas the product bound is 247566. Here the Smith factors are 11, 22506.

6.4.1 A nonsingular level-fifteen Hurwitz example

The universal integral basis at level fifteen is, in the chosen order,

$$[0], [2/5], [3/5], [4/5], [2/3], [1/15], [4/15], [7/15].$$

Corollary 6.9 (Explicit reductions at spectral order two). *Write $Z_r = \zeta(2, r/15)$ for $1 \leq r < 15$, and $Z_0 = \zeta(2)$. Then*

$$\zeta(2, 1/5) = \frac{729(Z_1 + Z_{11}) + 81(Z_2 + Z_7) + 9(Z_4 + Z_{14}) + (Z_8 + Z_{13})}{6560}, \quad (6.19)$$

$$\zeta(2, 1/3) = \frac{25(Z_1 + Z_4 + Z_7 + Z_{13}) + (Z_2 + Z_8 + Z_{11} + Z_{14})}{624}, \quad (6.20)$$

$$\zeta(2) = \frac{Z_1 + Z_2 + Z_4 + Z_7 + Z_8 + Z_{11} + Z_{13} + Z_{14}}{192}. \quad (6.21)$$

In the primitive order $(1, 2, 4, 7, 8, 11, 13, 14)/15$, the first two coefficient tuples are respectively

$$\frac{1}{6560}(729, 81, 9, 81, 1, 729, 1, 9), \quad \frac{1}{624}(25, 1, 25, 25, 1, 1, 25, 1).$$

The least common denominator of the full level-fifteen extension matrix is 511680, its primitive integral index is 4093440, and its nontrivial Smith factors are 2, 4, 511680.

Proof. Hurwitz multiplication uses the weights $t_p = p^s$, so here $a = 9$ and $b = 25$. The same permutation cycles used in the primitive-cokernel proof give an elementary elimination derivation of the formulas. Put $F_j = \zeta(2, j/5)$, and use multiplication by three. The four relations, in cycle order 1, 2, 4, 3, are

$$\begin{aligned} 9F_1 &= Z_1 + Z_{11} + F_2, & 9F_2 &= Z_2 + Z_7 + F_4, \\ 9F_4 &= Z_4 + Z_{14} + F_3, & 9F_3 &= Z_8 + Z_{13} + F_1. \end{aligned}$$

Eliminating F_2, F_4, F_3 gives (6.19), since $9^4 - 1 = 6560$. Multiplication by five gives the two-cycle

$$\begin{aligned} 25\zeta(2, 1/3) &= Z_1 + Z_4 + Z_7 + Z_{13} + \zeta(2, 2/3), \\ 25\zeta(2, 2/3) &= Z_2 + Z_8 + Z_{11} + Z_{14} + \zeta(2, 1/3). \end{aligned}$$

Elimination gives (6.20), since $25^2 - 1 = 624$. Finally, sorting the absolutely convergent Dirichlet series by residues prime to fifteen gives

$$\sum_{u \in U(15)} \zeta(2, u/15) = 15^2(1 - 3^{-2})(1 - 5^{-2})\zeta(2) = 192\zeta(2).$$

This proves the endpoint formula.

The cokernel parameters are $S_3 = 26$, $S_5 = 820$, $D_3 = 624$, $D_5 = 6560$, $g_1 = 2$, and $g_2 = 8$. Therefore the three nonunit Smith factors are 2, 4, $(624 \cdot 6560)/8 = 511680$ and the index is $624 \cdot 6560 = 4093440$. Equivalently, the exact denominator is

$$\text{lcm}(624, 6560, 192) = 511680.$$

The displayed formulas themselves show why every factor in this least common multiple is necessary: their coefficient lists contain $1/6560$, $1/624$, and $1/192$ in reduced form. The theorem controls all the other rows at once. Every zeta value in this example is finite; no Laurent finite-part convention is needed. $\square$

6.5 Research directions and integration scope

The following questions are natural next steps.

1. Determine the integral primitive-grid cokernel at three or more prime factors. The permutation-cycle calculation at two primes suggests a finite incidence presentation, but the resulting torsion need not be read from separate Euler factors.
2. Extend the explicit cokernel classification to prime powers $p^a \ell^b$ at ordinary and spectral weights. The quotient basis and resolution above already apply, but the primitive cokernel contains additional layers; for example, at $q = 12$ and ordinary weights its nontrivial torsion is $(\mathbb{Z}/2\mathbb{Z})^2$.
3. Give exact common denominators for arbitrary q , replacing the available product bound by an invariant-factor formula or a direct finite residue-lattice criterion.
4. Study reflection quotients over rings in which two is not a unit. The complex parity decomposition cannot simply be reduced modulo two; the integral basis makes the resulting presentation explicit without presupposing a decomposition.
5. Implement integral jet algebras, including characteristic- r truncations, on this same basis. The underlying quotient is free, whereas primitive coordinate inclusions can exhibit modular torsion as well as spectral vanishing.

These results refine the formal algebra of the manuscript. They do not change its proved complex rank formulas and do not establish arithmetic independence of polylogarithm, zeta, or Stieltjes values. The source sections inspected for this part contain no identified counterexample to their stated complex-algebra theorems. Their coefficient-ring hypothesis is essential to the character proof; Theorem 6.1 supplies a different argument when that hypothesis is removed.

7 Reconciling the two weight-seven Gaussian candidates

The pinned manuscript's Conjecture `cycloquot:conj:S6` and the incoming Cayley continuation state different-looking formulas for S_6 . They are exactly equivalent. This fact neither proves nor refutes the common conjecture, but it prevents the same proposed identity from being counted twice and removes an unnecessary product from one presentation. Write $G = \beta(2)$ and $L = \log 2$, and recall

$$S_6 = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^6}, \quad g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1).$$

The canonical candidate is

$$\begin{aligned} S_6 = & \frac{722}{527} g_{6,1} + \frac{40}{527} g_{4,3} - \frac{128}{155} g_{2,5} + \frac{15191}{28569600} \pi^7 \\ & - \frac{3}{124} G\zeta(5) - \frac{2373}{10540} \beta(4)\zeta(3) - 2\beta(6)L. \end{aligned} \tag{7.1}$$

It was supported in the baseline by two independent high-precision methods after its coefficients were frozen. Those numerical records are inherited evidence, not a new verification performed here. The present exact calculation concerns the equivalence of (7.1) and the incoming Cayley presentation [4].

Proposition 7.1 (Exact change of Gaussian coordinates). *With $g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1)$,*

$$g_{2,5} = 30g_{6,1} + 15g_{5,2} + 5g_{4,3} - \frac{11\pi^7}{368640} - \frac{15}{512} G\zeta(5) + \frac{3}{16} \beta(4)\zeta(3). \tag{7.2}$$

Consequently the manuscript's S_6 conjecture is equivalent to

$$S_6 \stackrel{?}{=} -\frac{12334}{527}g_{6,1} - \frac{384}{31}g_{5,2} - \frac{2136}{527}g_{4,3} + \frac{3179}{5713920}\pi^7 - \frac{801}{2108}\beta(4)\zeta(3) - 2\beta(6)L. \quad (7.3)$$

The latter is precisely the incoming Cayley candidate after dividing its stored primitive integer vector by its S_6 coefficient.

Proof. The convergent integral-shuffle products, valid first for $|z| < 1$ and then by ordinary convergence at $z = i$, are

$$\begin{aligned} \text{Li}_2(z) \text{Li}_5(z) &= 10F_{6,1}(z) + 5F_{5,2}(z) + 3F_{4,3}(z) + 2F_{3,4}(z) + F_{2,5}(z), \\ \text{Li}_3(z) \text{Li}_4(z) &= 20F_{6,1}(z) + 10F_{5,2}(z) + 4F_{4,3}(z) + F_{3,4}(z), \end{aligned}$$

where $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$. Subtract twice the second equation from the first. Since

$$\begin{aligned} \text{Im}(\text{Li}_2(i) \text{Li}_5(i)) &= -\frac{5\pi^7}{73728} - \frac{15}{512}G\zeta(5), \\ \text{Im}(\text{Li}_3(i) \text{Li}_4(i)) &= -\frac{7\pi^7}{368640} - \frac{3}{32}\beta(4)\zeta(3), \end{aligned}$$

taking imaginary parts gives (7.2). Substitution in the manuscript candidate cancels its entire $G\zeta(5)$ coefficient and gives (7.3).

For an explicit relation certificate, let R_{old} and R_{Cayley} denote the two residuals normalized to coefficient +1 on S_6 . Let $E_{p,q}$ denote the shuffle sum minus $\text{Im}(\text{Li}_p(i) \text{Li}_q(i))$. Then, as an exact formal coefficient identity,

$$R_{\text{old}} - R_{\text{Cayley}} = \frac{128}{155}(E_{2,5} - 2E_{3,4}).$$

The right-hand side vanishes by two convergent shuffles. The dependency-free program `verify_s6_equivalence.py` rebuilds the binomial shuffle rows and verifies this coefficient identity and the direct substitution independently, using exact rational arithmetic. $\square$

Scope of the search. A new exploratory system, within the broader standard-relation setting discussed in [11], was generated using the pinned incoming word convention. It contains 51 244 nonzero projected weight-seven double-shuffle rows and three convergent Cayley rows, with initial imaginary coordinates drawn from 24 514 possible convergent words. The double-shuffle inputs are convergent except for the permitted single factor $\text{Li}_1(1)$; its common divergence cancels between shuffle and stuffle. The Cayley seeds are the words of $F_{6,1}(i), F_{5,2}(i), F_{4,3}(i)$.

Sparse elimination modulo 1 000 003 was stopped for resource control after 8 000 logged pivots, while the target residual remained nonzero and fill-in had exceeded 20 million coefficient entries. This incomplete search is inconclusive: it proves neither membership nor nonmembership in the row span. It provides no proof of the S_6 conjecture, and the word “proved” should not be attached to either equivalent S_6 presentation. A future search should exploit shuffle indecomposables or a better sparse elimination ordering before expanding the entire weight-seven standard-relation space.

8 Proposed corrections and research-status updates

The audit found a logical overstatement in the rank-conjecture paragraph and a duplicated citation. The more substantial revisions concern research status and the range of validity of an argument. No counterexample was found to the stated fixed- m moment theorems or to the complex-algebra distribution theorems examined here. The following changes are proposed for editorial integration.

8.1 The rank-difference statement is a consequence

In `chapters/05-signed-kernels.tex`, for odd $w \geq 3$, the two conjectured ranks are

$$\text{rank } A_w = 5w - 6, \quad \text{rank } A_w^{-g} = \frac{9w - 11}{2}.$$

Let V be the row space and let π project onto the non-Gaussian columns. The exact linear-algebra identity is

$$\dim \ker(\pi|_V) = \text{rank } A_w - \text{rank } A_w^{-g}. \quad (8.1)$$

Thus the two rank formulas imply the stated Gaussian-supported dimension $(w - 1)/2$. That dimension alone fixes their difference, and does not determine both ranks separately. The word “Equivalently” should therefore be replaced by an implication, unless a separately established individual-rank statement is supplied. This correction is also recognized in the incoming rank work [2, 3]; it is carried forward here, not presented as a new mathematical discovery.

There is a more useful final editorial step than merely fixing this word: integrate one of the already supplied all-weight rank proofs, retain its exact matrix convention and certificate, and change the conjecture to a theorem with appropriate provenance. The incoming level-four article proves both odd-weight ranks and the even-weight formulas. The present article does not repeat that substantial proof.

8.2 A duplicate citation

The proof in `chapters/04-proportional-depth.tex` cites DLMF Section 2.3(iii) twice consecutively for the same invocation of Laplace’s method. One citation suffices. The accompanying `editorial/proposed_corrections.patch` makes this change and the minimal logical correction above against the pinned source. It deliberately leaves a conjecture labelled as such until its existing incoming proof is integrated.

8.3 The S_6 entry should have one identity and two coordinates

The two candidates reconciled in Section 7 should share one conjecture label, with (7.2) as the proved change of coordinates. Keep the numerical evidence attached to the candidate it actually tests, and cross-reference the equivalent form. The disappearance of $G\zeta(5)$ in the second presentation is an exact shuffle cancellation; it does not constitute independent evidence for a second relation. The incomplete modular search in this article provides neither a proof nor an obstruction.

8.4 Uniformity and integral-coordinate qualifications

The new harmonic theorem resolves the outstanding unbounded-ratio saddle problem, but its asymptotic parameter is still $p \rightarrow \infty$. Retain the fixed- p endpoint theorem. When interpreting the old large- α rate expansion, keep the actual first-pole condition $p/h - \log h \rightarrow +\infty$ distinct from merely $p/h \rightarrow \infty$.

The reflected-moment transition explicitly shows why one cannot substitute $m = m(n)$ into a fixed- m relative error without proof. This is a limitation already respected by the pinned theorem, not a contradiction to it. After integration, the remaining uniform problem should concern the bridges between the proved joint regimes and the interaction with optimal truncation when m also grows.

For distribution modules, retain the distinction between the free full quotient U_q and the image of its primitive grid. At level fifteen, ordinary weights give primitive rank seven over $\mathbb{Q}$ and five over $\mathbb{F}_2$, while U_{15} has rank eight over both fields. A character diagonalization which divides by a group order cannot simply be reduced modulo a prime dividing that order. The

integral presentation supplies the needed replacement. The baseline complex-algebra hypothesis was explicit and correct.

The spectral conventions also need to remain visible: Hurwitz multiplication has $t_p = p^s$, whereas polylogarithm multiplication has $t_p = p^{1-s}$. Thus the ordinary weight $t_p = 1$ corresponds respectively to $s = 0$ and $s = 1$. At a pole, combine the complete meromorphic identity before taking a limit; a coefficient that vanishes at the pole can still contribute a finite term.

9 Reproducibility and the meaning of the computations

The supplementary code is organized by theorem, with small machine-readable outputs beside the programs. Exact symbolic calculations are separated from floating-point quadrature. All finite checks are reproducible from the delivered archive; the large inconclusive S_6 search matrix is excluded.

9.1 Exact identities and finite algebra

- `code/harmonic/compile_inverse.py` regenerates the kernel and forward transition polynomials and derives Q_0, Q_1 by exact symbolic inversion. It verifies the two coefficients in the original normalized expression as well as the logarithmic calculation. `code/gaussian/audit_inverse_q1.py` independently starts from shifted gamma cumulants and returns the same result.
- The moment scripts derive P_1, P_2 twice: directly from the centered gamma expectation and from finite offset-residue identities. The results agree symbolically. The finite all-orders generator produces generic $P_0, \dots, P_3$ by default and verifies the degree bounds. These coefficients use symbols for the local analytic germs, so their dependence can be inspected before substituting gamma and zeta constants.
- `code/distribution/verify_integral_distribution.py` checks complete polynomial normal-form identities at every level $2 \leq q \leq 60$, nineteen symbolic primitive determinants, 112 finite-characteristic resolutions, and 504 complete two-prime Smith specializations. It checks the three displayed level-fifteen coefficient tuples over $\mathbb{Q}$. These finite tests check an implementation; the all-level and arbitrary-ring assertions follow from the finite-resolution and cycle proofs.
- `code/gaussian/verify_s6_equivalence.py` uses only Python's integer and rational arithmetic. It reconstructs the two binomial shuffle rows, verifies the exact residual combination, and verifies direct elimination of $g_{2,5}$. It does not evaluate or prove the common S_6 conjecture.

9.2 The exact interval step in the slope-ratio theorem

The monotonicity theorem in Section 5 has a different computational status from the diagnostic tables. Its middle-interval inequality is a finite part of the proof. The program `code/moments/certify_gamma_saddle.py` uses only integers and rationals. It independently encloses γ , the required zeta values, and logarithms; bounds all differentiated Taylor tails; and evaluates entire rational subintervals with outward rounding. The article specifies every series, remainder and interval endpoint. The certificate covers the compact interval completely, and analytic estimates handle the two endpoints. A second inspection and replay produced the identical rational bound. This is an exact computer-assisted proof, not evidence inferred from sampled signs.

9.3 High-precision diagnostics

The harmonic diagnostics evaluate the exact positive integral after centering at its saddle, at fifty working decimal digits. The test set includes $p/h \rightarrow 0$, bounded ratio, very large ratio, increasing

shift, and transition points. The inverse calculation uses the same defining integral to locate the target fraction. The largest sampled value of $p^2|R/\mathcal{S} - 1 - E_1/p|$ is about 0.194; this is a sample statistic, not a universal error constant.

The joint-moment diagnostics use eighty working digits and fifteen pairs (n, m) near three transition offsets. They integrate the defining expression with $x = e^{-y}$, centered and standardized around the gamma mean. Two finite integration-window choices are compared. Their very small discrepancy is a consistency check; it does not by itself bound all quadrature or omitted-tail error. The proof’s localization estimates and Taylor bounds establish the asymptotic theorem independently of these diagnostics.

The level-fifteen Hurwitz example is also evaluated independently at ninety working digits, with a largest recorded absolute discrepancy below $2.46 \cdot 10^{-91}$. Its proof, however, is the exact distribution elimination, not the small decimal residual.

Figure 1 displays the reduction in error when successive corrections are retained. It is generated directly from the delivered diagnostic JSON files.

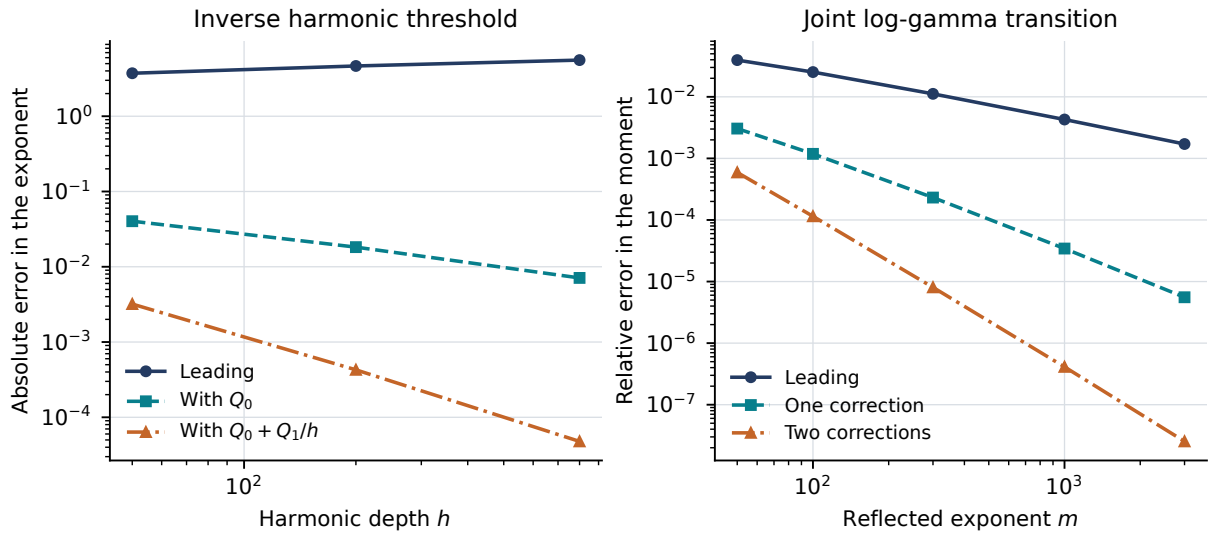


Figure 1: Successive asymptotic corrections in the inverse harmonic problem and the joint-moment transition. All plotted values are high-precision diagnostics of the defining integrals. They are distinct from the exact interval certificate used in the monotonicity proof.

9.4 Replay and portability

Python 3.10 or later suffices for the supplied core scripts. The symbolic and diagnostic programs additionally use SymPy and mpmath; the versions used in this delivery are recorded in `provenance/verification_receipt.json`. The package-level `verify.py` runs the exact checks in a temporary directory, compares the critical saved results, and records a fresh verification receipt. Numerical diagnostics are available as an optional replay, since they take longer and are not proof gates for the analytic theorems. The T_EX source builds with a standard recent TeX Live installation using `latexmk -pdf article.tex`.

The source receipts identify all thirty-nine inspected consolidated source files and the six incoming archives. SHA-256 checksums cover the delivered artifacts. These records make the scope of the audit and the origin of the baseline unambiguous, without redistributing the entire existing manuscript or its incoming packages.

10 Further research questions

The following programme prioritizes questions for which the present results supply a concrete starting point. Some are new refinements of the baseline questions; others remain difficult after the progress made here. The statements in this section are proposals, not additional theorems.

10.1 Harmonic sums and uniform asymptotics

- H1. **Certified relative remainders on the entire saddle range.** The proof of Theorem 2.1 supplies uniform derivative and curvature control, including the explicit bound (2.14). Derive useful computable constants C_J, p_J such that

$$\left| \frac{R_{p,h}}{\mathcal{S}_{p,h}} - \sum_{j=0}^J E_j p^{-j} \right| \leq C_J p^{-J-1} \quad (p \geq p_J)$$

holds for every $h \geq 1, a > 0$. The main task is to retain practical constants in the Gaussian Taylor and endpoint bounds.

- H2. **A bridge to bounded p .** The new theorem is uniform in p/h , but requires $p \rightarrow \infty$. Find a single approximation which joins it to the fixed- p , growing- h endpoint expansion. An incomplete-gamma or gamma-density formulation may be more natural than forcing a Gaussian approximation when its shape parameter stays bounded.
- H3. **Inverse fractions approaching zero or one.** The inverse theorem keeps θ away from both endpoints. Determine the largest moving ranges of $\lambda = -\log \theta$ for which the polynomial inverse remains uniform. For example, how do the scales change if $\lambda \rightarrow 0$ or $\lambda \rightarrow \infty$ with h ? Near the lower endpoint $(\log 2)^h/2$, the transition underlying the inverse itself changes.
- H4. **Late saddle coefficients.** The all-orders expansion is effective at every fixed order, but does not describe the growth of E_j as $j \rightarrow \infty$. Identify the complex saddles and singularities controlling that growth, and determine whether an optimal-truncation theorem can be uniform over the moving real saddle. This would connect the harmonic analysis with the manuscript's sharp reflected-moment work.

10.2 Two growing log-gamma exponents

- M1. **An analytic slope-ratio inequality and its sharp constant.** Theorem 5.1 proves global monotonicity using an exact interval certificate in the middle. Find a proof based entirely on analytic inequalities, and determine the optimal lower bound for J'/J . The symmetry suggests that the midpoint deserves special attention, but the certificate given here does not assert that it is the minimizer.
- M2. **Moving transition parameter.** Extend Theorem 4.1 to transition parameters which tend to zero or infinity. Its correction polynomials exhibit how powers of λ compete with powers of m^{-1} , but that formal observation is not a uniform bound. Determine the first range in which the cubic and higher local perturbations change the leading transition function.
- M3. **Uniform bridges between moment regimes.** Construct an approximation valid while the dominant saddle travels from a fixed interior point to $x \asymp m^{-1}$ and beyond. The transition limit proved here supplies a necessary boundary condition. A successful formula must also control the opposite endpoint, rather than using a local series alone.
- M4. **A general analytic endpoint-pair theorem.** For $f(x) = -\log x + g(x)$ and $B(x) = b_1 x + b_2 x^2 + \dots$, the local argument suggests the leading factor

$$\exp((g_1 + b_2/b_1)\lambda).$$

State and prove explicit global conditions under which this generalization is valid. The exact coefficient recipe indicates which parts are formal; the endpoint localization identifies where additional hypotheses are needed.

- M5. **Optimal truncation with a growing reflected exponent.** Determine the maximal growth of m for which the fixed- m half-first-omitted-term law remains uniform. The factor $B(-\tau)^m$ in the inverse-gamma amplitude changes the late-term saddle. A theorem must track this effect together with the transition polynomials, not simply substitute a variable m into a fixed-amplitude estimate.

10.3 Integral distribution lattices

- D1. **Three or more prime divisors.** Determine coker P_q integrally for squarefree q with at least three prime divisors. The two-prime core suggests an incidence presentation involving intersections of divisor layers. Which torsion classes require more than pairwise cycle information?
- D2. **Prime powers in the primitive cokernel.** Extend the explicit classification from $p\ell$ to $p^a\ell^b$. The full quotient already has a basis and finite resolution, so the difficulty is the primitive inclusion. Separate the effect of higher primary heights from multiplication-cycle torsion.
- D3. **An exact denominator formula for every level.** For $q = p\ell$ the answer is a least common multiple of three explicit factors. For general q , seek a formula for the exponent of the integral primitive cokernel, rather than merely its determinant. A product bound can exceed the needed denominator substantially, as the level-fifteen example demonstrates.
- D4. **Integral reflection and jet modules.** Study reflection quotients in characteristic two and spectral jets over integer or finite-characteristic coefficient rings. The full distribution module remains free, but the primitive image may simultaneously have spectral defects and modular torsion. A presentation that keeps both effects should be useful for special-value computations without semisimple character projectors.

10.4 Gaussian polylogarithms and exact discovery

- G1. **Prove or refute the common S_6 conjecture.** Use the reduced coordinate form (7.3). The weight-five success indicates that transformations of iterated-integral paths can supply information outside a restricted depth-two product module. A successful proof must display the extra analytic relation and an exact finite combination, not only a numerical null vector.
- G2. **Control relation-space growth before elimination.** The weight-seven experiment shows how sparse rows can become dense during elimination. Work first in shuffle indecomposables, exploit color and reversal symmetries, and isolate the relevant quotient before adjoining transformed paths. A modular calculation which suggests membership should be followed by rational reconstruction and exact verification of every relation.
- G3. **Obstructions with an explicit relation vocabulary.** Extend the existing uniform depth-two obstructions to carefully specified larger presentations. A nonzero functional on a restricted module is useful because it identifies missing information; it is not a theorem about all functional equations of the evaluated periods. State the alphabet, regularization rules, allowed depths, and parameter transformations in each obstruction theorem.
- G4. **A reproducible conjecture registry.** Store each candidate with an immutable coordinate order, a primitive integer relation, discovery precision, frozen-coefficient reevaluation, and its exact relations to other candidates. The two S_6 presentations show why coordinate

equivalence should be checked before counting independent discoveries. Keep numerical evidence, formal proofs, and interval certificates as distinct fields.

These questions preserve the central aim of the programme: explain why an identity holds, identify the structure that makes it computable, and state precisely where an argument ceases to be uniform or a formal relation ceases to determine an arithmetic one.

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